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International vocabulary of metrology — Basic and general concepts and associated terms (VIM)

Third edition

Joint Committee for Guides in Metrology

JCGM
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International vocabulary of metrology — Basic and general concepts and associated terms (VIM)

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Foreword

In 1997 the Joint Committee for Guides in Metrology (JCGM), chaired by the Director of the BIPM, was formed by the seven International Organizations that had prepared the original versions of the *Guide to the expression of uncertainty in measurement (GUM)* and the *International vocabulary of basic and general terms in metrology (VIM)* . The Joint Committee took on this part of the work of the ISO Technical Advisory Group 4 (TAG 4), which had developed the GUM and the VIM. The Joint Committee was originally made up of representatives from the International Bureau of Weights and Measures (BIPM), the International Electrotechnical Commission (IEC), the International Federation of Clinical Chemistry and Laboratory Medicine (IFCC), the International Organization for Standardization (ISO), the International Union of Pure and Applied Chemistry (IUPAC), the International Union of Pure and Applied Physics (IUPAP), and the International Organization of Legal Metrology (OIML). In 2005, the International Laboratory Accreditation Cooperation (ILAC) officially joined the seven founding international organizations.

The JCGM has two Working Groups. Working Group 1 (JCGM/WG 1) on the GUM has the task of promoting the use of the GUM and preparing Supplements to the GUM for broad application. Working Group 2 (JCGM/WG 2) on the VIM has the task of revising the VIM and promoting its use. Working Group 2 is composed of up to two representatives of each member organization, supplemented by a limited number of experts. The third edition of the VIM has been prepared by Working Group 2.

In 2004, a first draft of the third edition of the VIM was submitted for comments and proposals to the eight organizations represented in the JCGM, which in most cases consulted their members or affiliates, including numerous National Metrology Institutes. Comments were studied and discussed, taken into account when appropriate, and replied to by JCGM/WG 2. A final draft of the third edition was submitted in 2006 to the eight organizations for review and approval.

All subsequent comments were considered and taken into account as appropriate by Working Group 2.

The third edition of the VIM has been approved by each and all of the eight JCGM Member organizations.

This third edition cancels and replaces the second edition 1993.

Introduction

In general, a vocabulary is a "terminological dictionary which contains designations and definitions from one or more specific subject fields" ([ISO 1087-1:2000], 3.7.2). The present Vocabulary pertains to metrology, the "science of measurement and its application". It also covers the basic principles governing quantities and units. The field of quantities and units could be treated in many different ways. Clause 1 of this Vocabulary is one such treatment, and is based on the principles laid down in the various parts of ISO 31, *Quantities and units* , currently being replaced by ISO 80000 and IEC 80000 series *Quantities and units*, and in theSI Brochure, *The International System of Units* (published by the BIPM).

The second edition of the *International vocabulary of basic and general terms in metrology* (VIM) was published in 1993. The need to cover measurements in chemistry and laboratory medicine for the first time, as well as to incorporate concepts such as those that relate to metrological traceability, measurement uncertainty, and nominal properties, led to this third edition. Its title is now *International vocabulary of metrology—Basic and general concepts and associated terms (VIM)* , in order to emphasize the primary role of concepts in developing a vocabulary.

In this Vocabulary, it is taken for granted that there is no fundamental difference in the basic principles of measurement in physics, chemistry, laboratory medicine, biology, or engineering. Furthermore, an attempt has been made to meet conceptual needs of measurement in fields such as biochemistry, food science, forensic science, and molecular biology.

Several concepts that appeared in the second edition of the VIM do not appear in this third edition because they are no longer considered to be basic or general. For example, the concept 'response time', used in describing the temporal behaviour of a measuring system, is not included. For concepts related to measurement devices that are not covered by this third edition of the VIM, the reader should consult other vocabularies such as IEC 60050, *International Electrotechnical Vocabulary* , IEV. For concepts concerned with quality management, mutual recognition arrangements pertaining to metrology, or legal metrology, the reader is referred to documents given in the bibliography.

Development of this third edition of the VIM has raised some fundamental questions about different current philosophies and descriptions of measurement, as will be summarized below. These differences sometimes lead to difficulties in developing definitions that could be used across the different descriptions. No preference is given in this third edition to any of the particular approaches.

The change in the treatment of measurement uncertainty from an Error Approach (sometimes called Traditional Approach or True Value Approach) to an Uncertainty Approach necessitated reconsideration of some of the related concepts appearing in the second edition of the VIM. The objective of measurement in the Error Approach is to determine an estimate of the true value that is as close as possible to that single true value. The deviation from the true value is composed of random and systematic errors. The two kinds of errors, assumed to be always distinguishable, have to be treated differently. No rule can be derived on how they combine to form the total error of any given measurement result, usually taken as the estimate. Usually, only an upper limit of the absolute value of the total error is estimated, sometimes loosely named "uncertainty".

In the CIPM Recommendation INC-1 (1980) on the Statement of Uncertainties, it is suggested that the components of measurement uncertainty should be grouped into two categories, Type A and Type B, according to whether they were evaluated by statistical methods or otherwise, and that they be combined to yield a variance according to the rules of mathematical probability theory by also treating the Type B components in terms of variances. The resulting standard deviation is an expression of a measurement uncertainty. A view of the Uncertainty Approach was detailed in the *Guide to the expression of uncertainty in measurement (GUM)* (1993, corrected and reprinted in 1995) that focused on the mathematical treatment of measurement uncertainty through an explicit measurement model under the assumption that the measurand can be characterized by an essentially unique value. Moreover, in the GUM as well as in IEC documents, guidance is provided on the Uncertainty Approach in the case of a single reading of a calibrated instrument, a situation normally met in industrial metrology.

The objective of measurement in the Uncertainty Approach is not to determine a true value as closely as possible. Rather, it is assumed that the information from measurement only permits assignment of an interval of reasonable values to the measurand, based on the assumption that no mistakes have been made in performing the measurement.

Additional relevant information may reduce the range of the interval of values that can reasonably be attributed to the measurand. However, even the most refined measurement cannot reduce the interval to a single value because of the finite amount of detail in the definition of a measurand. The definitional uncertainty, therefore, sets a minimum limit to any measurement uncertainty. The interval can be represented by one of its values, called a “measured quantity value”.

In the GUM, the definitional uncertainty is considered to be negligible with respect to the other components of measurement uncertainty. The objective of measurement is then to establish a probability that this essentially unique value lies within an interval of measured quantity values, based on the information available from measurement.

The IEC scenario focuses on measurements with single readings, permitting the investigation of whether quantities vary in time by demonstrating whether measurement results are compatible. The IEC view also allows non-negligible definitional uncertainties. The validity of the measurement results is highly dependent on the metrological properties of the instrument as demonstrated by its calibration. The interval of values offered to describe the measurand is the interval of values of measurement standards that would have given the same indications.

In the GUM, the concept of true value is kept for describing the objective of measurement, but the adjective “true” is considered to be redundant. The IEC does not use the concept to describe this objective. In this Vocabulary, the concept and term are retained because of common usage and the importance of the concept.

Conventions

Terminology rules

The definitions and terms given in this third edition, as well as their formats, comply as far as possible with the rules of terminology work, as outlined in [ISO 704:2000, ISO 1087-1:2000] and [ISO 10241:1992]. In particular, the substitution principle applies; that is, it is possible in any definition to replace a term referring to a concept defined elsewhere in the VIM by the definition corresponding to that term, without introducing contradiction or circularity.

Concepts are listed in five chapters and in logical order in each chapter.

In some definitions, the use of non-defined concepts (also called "primitives") is unavoidable. In this Vocabulary, such non-defined concepts include: system, component, phenomenon, body, substance, property, reference, experiment, examination, magnitude, material, device, and signal.

To facilitate the understanding of the different relations between the various concepts given in this Vocabulary, concept diagrams have been introduced. They are given in Annex A.

Reference number

Concepts appearing in both the second and third editions have a double reference number; the third edition reference number is printed in bold face, and the earlier reference from the second edition is given in parentheses and in light font.

Synonyms

Multiple terms for the same concept are permitted. If more than one term is given, the first term is the preferred one, and it is used throughout as far as possible.

Bold face

Terms used for a concept to be defined are printed in **bold face**. In the text of a given entry, terms of concepts defined elsewhere in the VIM are also printed in **bold face** the first time they appear.

Quotation marks

In the English text of this document, single quotation marks ('…') surround the term representing a concept unless it is in bold. Double quotation marks ("…") are used when only the term is considered, or for a quotation. In the French text, quotation marks («…») are used for quotations, or to highlight a word or a group of words.

Decimal sign

The decimal sign in the English text is the point on the line, and the comma on the line is the decimal sign in the French text.

French terms "mesure" and "mesurage" ("measurement")

The French word "mesure" has several meanings in everyday French language. For this reason, it is not used in this Vocabulary without further qualification. It is for the same reason that the French word "mesurage" has been introduced to describe the act of measurement. Nevertheless, the French word "mesure" occurs many times in forming terms in this Vocabulary, following current usage, and without ambiguity. Examples are: instrument de mesure, appareil de mesure, unité de mesure, méthode de mesure. This does not mean that the use of the French word “mesurage” in place of “mesure” in such terms is not permissible when advantageous.

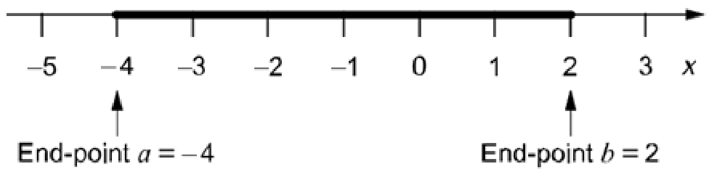
Equal-by-definition symbol

The symbol := denotes “is by definition equal to” as given in the ISO 80000 and IEC 80000 series.

Interval

The term “interval” is used together with the symbol $[a, b]$ to denote the set of real numbers x for which $a \leq x \leq b$, where a and b a are real numbers. The term “interval” is used here for ‘closed interval’. The symbols a and b denote the ‘end-points’ of the interval $[a, b]$.

EXAMPLE $[-4, 2]$



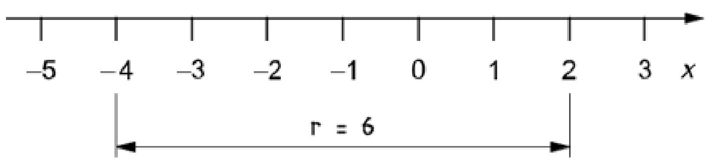
The two end-points 2 and -4 of the interval $[4, 2]$ can be stated as -1 3. The latter expression does not denote the interval $[-4, 2]$. Nevertheless, -1 3 is often used to denote the interval $[-4, 2]$.

Range of interval

Range

The range of the interval $[a, b]$ is the difference b a and is denoted by $r[a, b]$.

EXAMPLE $r[-4, 2] = 2 - (-4) = 6$



NOTE: The term “span” is sometimes used for this concept.

Scope

In this Vocabulary, a set of definitions and associated terms is given, in English and French, for a system of basic and general concepts used in metrology, together with concept diagrams to demonstrate their relations. Additional information is given in the form of examples and notes under many definitions.

This Vocabulary is meant to be a common reference for scientists and engineers — including physicists, chemists, medical scientists — as well as for both teachers and practitioners involved in planning or performing measurements, irrespective of the level of measurement uncertainty and irrespective of the field of application. It is also meant to be a reference for governmental and inter-governmental bodies, trade associations, accreditation bodies, regulators, and professional societies.

Concepts used in different approaches to describing measurement are presented together. The member organizations of the JCGM can select the concepts and definitions in accordance with their respective terminologies. Nevertheless, this Vocabulary is intended to promote global harmonization of terminology used in metrology.

International vocabulary of metrology — Part : Basic and general concepts and associated terms (VIM)

3rd edition

1. Quantities and units

For the purposes of this document, the following terms and definitions apply.

1.1.
quantity
property of a phenomenon, body, or substance, where the property has a magnitude that can be expressed as a number and a reference

Note 1 to entry: The generic concept ‘quantity’ can be divided into several levels of specific concepts, as shown in the following table. The left hand side of the table shows specific concepts under ‘quantity’. These are generic concepts for the individual quantities in the right hand column.

length, l longueur, l	radius, r rayon, r	radius of circle A, rA or $r(\mathrm{A})$ rayon du cercle A, rA or $r(\mathrm{A})$
	wavelength, longueur d’onde,	wavelength of the sodium D radiation, ${}_{\mathrm{D}}$ or (D; Na) longueur d’onde de la radiation D du sodium, ${}_{\mathrm{D}}$ ou (D; Na)
energy, E énergie, E	kinetic energy, T énergie cinétique, T	kinetic energy of particle i in a given system, T_i énergie cinétique de la particule i dans un système donné, T_i
	heat, Q chaleur, Q	heat of vaporization of sample i of water, Q_i chaleur de vaporisation du spécimen i d’eau, Q_i
electric charge, Q charge électrique, Q		electric charge of the proton, e charge électrique du proton, e
electric resistance, R résistance électrique, R		electric resistance of resistor i in a given circuit, R_i résistance électrique de la résistance i dans un circuit donné, R_i
amount-of-substance concentration of entity B, c_{B} concentration en quantité de matière du constituant B, c_{B}		amount-of-substance concentration of ethanol in wine sample i, c_i (C_2H_5OH) concentration en quantité de matière d’éthanol dans le spécimen i de vin, c_i (C_2H_5OH)

number concentration of entity B, C_B nombre volumique du constituant B, C_B	number concentration of erythrocytes in blood sample *i*, C (Erys; B_i) nombre volumique d'érythrocytes dans le spécimen *i* de sang, C (Erc; Sg_i)
Rockwell C hardness, HRC dureté C de Rockwell, HRC	Rockwell C hardness of steel sample *i*, HRC_i dureté C de Rockwell du spécimen *i* d'acier, HRC_i

Note 2 to entry: A reference can be a **measurement unit**, a **measurement procedure**, a **reference material**, or a combination of such.

Note 3 to entry: Symbols for quantities are given in the ISO 80000 and IEC 80000 series *Quantities and units*. The symbols for quantities are written in italics. A given symbol can indicate different quantities.

Note 4 to entry: The preferred IUPAC-IFCC format for designations of quantities in laboratory medicine is "System-Component; kind-of-quantity".

Note 5 to entry: A quantity as defined here is a scalar. However, a vector or a tensor, the components of which are quantities, is also considered to be a quantity.

Note 6 to entry: The concept 'quantity' may be generically divided into, e.g. 'physical quantity', 'chemical quantity', and 'biological quantity', or **base quantity** and **derived quantity**.

EXAMPLE "Plasma (Blood) Sodium ion; amount-of-substance concentration equal to 143 mol · l^{-1} in a given person at a given time".

1.2.
(1.1, Note 2) kind of quantity
kind

aspect common to mutually comparable **quantities**

Note 1 to entry: The division of 'quantity' according to 'kind of quantity' is to some extent arbitrary.

Note 2 to entry: Quantities of the same kind within a given **system of quantities** have the same **quantity dimension**. However, quantities of the same dimension are not necessarily of the same kind.

Note 3 to entry: In English, the terms for quantities in the left half of the table in 1.1, Note 1, are often used for the corresponding 'kinds of quantity'. In French, the term "nature" is only used in expressions such as "grandeurs de même nature" (in English, "quantities of the same kind").

EXAMPLE 1 The quantities diameter, circumference, and wavelength are generally considered to be quantities of the same kind, namely of the kind of quantity called length.

EXAMPLE 2 The quantities heat, kinetic energy, and potential energy are generally considered to be quantities of the same kind, namely of the kind of quantity called energy.

EXAMPLE 3 The quantities moment of force and energy are, by convention, not regarded as being of the same kind, although they have the same dimension. Similarly for heat capacity and entropy, as well as for number of entities, relative permeability, and mass fraction.

1.3.
(1.2) system of quantities
set of **quantities** together with a set of non-contradictory equations relating those quantities

Note 1 to entry: **Ordinal quantities**, such as Rockwell C hardness, are usually not considered to be part of a system of quantities because they are related to other quantities through empirical relations only.

1.4.
(1.3) base quantity
quantity in a conventionally chosen subset of a given **system of quantities**, where no subset quantity can be expressed in terms of the others

Note 1 to entry: The subset mentioned in the definition is termed the “set of base quantities”.

Note 2 to entry: Base quantities are referred to as being mutually independent since a base quantity cannot be expressed as a product of powers of the other base quantities.

Note 3 to entry: ‘Number of entities’ can be regarded as a base quantity in any system of quantities.

EXAMPLE The set of base quantities in the **International System of Quantities (ISQ)** is given in 1.6.

1.5.
(1.4) derived quantity
quantity, in a **system of quantities**, defined in terms of the **base quantities** of that system

EXAMPLE In a system of quantities having the base quantities length and mass, mass density is a derived quantity defined as the quotient of mass and volume (length to the third power).

1.6.
International System of Quantities
ISQ

system of quantities based on the seven **base quantities**: length, mass, time, electric current, thermodynamic temperature, amount of substance, and luminous intensity

Note 1 to entry: This system of quantities is published in the ISO 80000 and IEC 80000 series *Quantities and units*.

Note 2 to entry: The **International System of Units (SI)** (see 1.16) is based on the ISQ.

1.7.
(1.5) quantity dimension
dimension of a quantity
dimension

expression of the dependence of a **quantity** on the **base quantities** of a **system of quantities** as a product of powers of factors corresponding to the base quantities, omitting any numerical factor
Thus, the dimension of a quantity Q is denoted by $\dim Q = \mathsf{L}\ \mathsf{M}\ \mathsf{T}\ \mathsf{I}\ \Theta\ \mathsf{N}\ \mathsf{J}$ where the exponents, named dimensional exponents, are positive, negative, or zero.

Note 1 to entry: A power of a factor is the factor raised to an exponent. Each factor is the dimension of a base quantity.

Note 2 to entry: The conventional symbolic representation of the dimension of a base quantity is a single upper case letter in roman (upright) sans-serif type. The conventional symbolic representation of the dimension of a **derived quantity** is the product of powers of the dimensions of the base quantities according to the definition of the derived quantity. The dimension of a quantity Q is denoted by dim Q.

Note 3 to entry: In deriving the dimension of a quantity, no account is taken of its scalar, vector, or tensor character.

Note 4 to entry: In a given system of quantities,

- quantities of the same **kind** have the same quantity dimension,
- quantities of different quantity dimensions are always of different kinds, and
- quantities having the same quantity dimension are not necessarily of the same kind.

Note 5 to entry: Symbols representing the dimensions of the base quantities in the ISQ are:

Base quantity Grandeur de base	Symbol for dimension Symbole de la dimension
length longueur	L
mass masse	M
time temps	T
electric current courant électrique	I
thermodynamic temperature température thermodynamique	Θ
amount of substance quantité de matière	N
luminous intensity intensité lumineuse	J

EXAMPLE 1 In the **ISQ**, the quantity dimension of force is denoted by $\dim F = \mathrm{LMT}^{-2}$.

EXAMPLE 2 In the same system of quantities, $\dim \rho B = \mathrm{ML}^{-3}$ is the quantity dimension of mass concentration of component B, and ML^{-3} is also the quantity dimension of mass density, ρ, (volumic mass).

EXAMPLE 3 The period T of a pendulum of length l at a place with the local acceleration of free fall g is

$$\mathrm{T} = 2\pi\sqrt{\frac{l}{g}} \text{ or } \mathrm{T} = \mathrm{C}(g)\sqrt{l}$$

where $\mathrm{C}(g) = \frac{2\pi}{\sqrt{g}}$ Hence $\dim \mathrm{C}(g) = \mathrm{L}^{-1/2}\mathrm{T}$.

1.8.
(1.6) quantity of dimension one
dimensionless quantity

quantity for which all the exponents of the factors corresponding to the **base quantities** in its **quantity dimension** are zero

Note 1 to entry: The term “dimensionless quantity” is commonly used and is kept here for historical reasons. It stems from the fact that all exponents are zero in the symbolic representation of the dimension for such quantities. The term “quantity of dimension one” reflects the convention in which the symbolic representation of the dimension for such quantities is the symbol 1 (see [ISO 31-0:1992], 2.2.6).

Note 2 to entry: The **measurement units** and **values** of quantities of dimension one are numbers, but such quantities convey more information than a number.

Note 3 to entry: Some quantities of dimension one are defined as the ratios of two quantities of the same **kind**.

Note 4 to entry: Numbers of entities are quantities of dimension one.

EXAMPLE 1 Plane angle, solid angle, refractive index, relative permeability, mass fraction, friction factor, Mach number.

EXAMPLE 2 Number of turns in a coil, number of molecules in a given sample, degeneracy of the energy levels of a quantum system.

1.9.
(1.7) measurement unit
unit of measurement
unit

real scalar **quantity**, defined and adopted by convention, with which any other quantity of the same **kind** can be compared to express the ratio of the two quantities as a number

Note 1 to entry: Measurement units are designated by conventionally assigned names and symbols.

Note 2 to entry: Measurement units of quantities of the same **quantity dimension** may be designated by the same name and symbol even when the quantities are not of the same kind. For example, joule per kelvin and $J \cdot K^{-1}$ are respectively the name and symbol of both a measurement unit of heat capacity and a measurement unit of entropy, which are generally not considered to be quantities of the same kind. However, in some cases special measurement unit names are restricted to be used with quantities of a specific kind only. For example, the measurement unit 'second to the power minus one' ($1 \cdot s^{-1}$) is called hertz (Hz) when used for frequencies and becquerel (Bq) when used for activities of radio nuclides.

Note 3 to entry: Measurement units of **quantities of dimension one** are numbers. In some cases these measurement units are given special names, e.g. radian, steradian, and decibel, or are expressed by quotients such as millimole per mole equal to 10^{-3} and microgram per kilogram equal to 10^{-9}.

Note 4 to entry: For a given quantity, the short term "unit" is often combined with the quantity name, such as "mass unit" or "unit of mass".

1.10.
(1.13) base unit
measurement unit that is adopted by convention for a **base quantity**

Note 1 to entry: In each **coherent system of units**, there is only one base unit for each base quantity.

Note 2 to entry: A base unit may also serve for a **derived quantity** of the same **quantity dimension**.

Note 3 to entry: For number of entities, the number one, symbol 1, can be regarded as a base unit in any **system of units**.

EXAMPLE 1 In the **SI**, the metre is the base unit of length. In the CGS systems, the centimetre is the base unit of length.

EXAMPLE 2 Rainfall, when defined as areic volume (volume per area), has the metre as a **coherent derived unit** in the SI.

1.11.
(1.14) derived unit
measurement unit for a **derived quantity**

EXAMPLE The metre per second, symbol $m \cdot s^{-1}$, and the centimetre per second, symbol $cm \cdot s^{-1}$, are derived units of speed in the **SI**. The kilometre per hour, symbol $km \cdot h^{-1}$, is a measurement unit of speed outside the SI but accepted for use with the SI. The knot, equal to one nautical mile per hour, is a measurement unit of speed outside the SI.

1.12.
(1.10) coherent derived unit
derived unit that, for a given **system of quantities** and for a chosen set of **base units**, is a product of powers of base units with no other proportionality factor than one

Note 1 to entry: A power of a base unit is the base unit raised to an exponent.

Note 2 to entry: Coherence can be determined only with respect to a particular system of quantities and a given set of base units.

Note 3 to entry: A derived unit can be coherent with respect to one system of quantities but not to another.

Note 4 to entry: The coherent derived unit for every derived **quantity of dimension one** in a given system of units is the number one, symbol 1. The name and symbol of the **measurement unit** one are generally not indicated.

EXAMPLE 1 If the metre, the second, and the mole are base units, the metre per second is the coherent derived unit of velocity when velocity is defined by the **quantity equation** $v = d\mathbf{r} / dt$, and the mole per cubic metre is the coherent derived unit of amount-of-substance concentration when amount-of-substance concentration is defined by the quantity equation $c = n / V$. The kilometre per hour and the knot, given as examples of derived units in 1.11, are not coherent derived units in such a system of quantities.

EXAMPLE 2 The centimetre per second is the coherent derived unit of speed in a CGS **system of units** but is not a coherent derived unit in the **SI**.

1.13.
(1.9) system of units
set of **base units** and **derived units**, together with their multiples and submultiples, defined in accordance with given rules, for a given **system of quantities**

1.14.
(1.11) coherent system of units
system of units, based on a given **system of quantities**, in which the **measurement unit** for each **derived quantity** is a **coherent derived unit**

Note 1 to entry: A system of units can be coherent only with respect to a system of quantities and the adopted **base units**.

Note 2 to entry: For a coherent system of units, **numerical value equations** have the same form, including numerical factors, as the corresponding **quantity equations**.

EXAMPLE Set of coherent SI units and relations between them.

1.15.
(1.15) off-system measurement unit
off-system unit

measurement unit that does not belong to a given **system of units**

EXAMPLE 1 The electron-volt (about $1.60218 \cdot 10^{-19}$ J) is an off-system measurement unit of energy with respect to the **SI**.

EXAMPLE 2 Day, hour, minute are off-system measurement units of time with respect to the SI.

1.16.
(1.12) International System of Units
SI

system of units, based on the **International System of Quantities**, their names and symbols, including a series of prefixes and their names and symbols, together with rules for their use, adopted by the General Conference on Weights and Measures (CGPM)

Note 1 to entry: The SI is founded on the seven **base quantities** of the **ISQ** and the names and symbols of the corresponding **base units** that are contained in the following table.

Base quantity **Grandeur de base**	**Base unit** **Unité de base**	
Name **Nom**	**Name** **Nom**	**Symbol** **Symbole**
length longueur	metre mètre	m
mass masse	kilogram kilogramme	kg
time temps	second seconde	s

electric current courant électrique	ampere ampère	A
thermodynamic temperature température thermodynamique	kelvin kelvin	K
amount of substance quantité de matière	mole mole	mol
luminous intensity intensité lumineuse	candela candela	cd

Note 2 to entry: The base units and the **coherent derived units** of the SI form a coherent set, designated the "set of coherent SI units".

Note 3 to entry: For a full description and explanation of the International System of Units, see the current edition of the SI brochure published by the Bureau International des Poids et Mesures (BIPM) and available on the BIPM website.

Note 4 to entry: In **quantity calculus**, the quantity 'number of entities' is often considered to be a base quantity, with the base unit one, symbol 1.

Note 5 to entry: The SI prefixes for **multiples of units** and **submultiples of units** are:

Factor **Facteur**	**Prefix** **Préfixe**	
	Name **Nom**	**Symbol** **Symbole**
10^{24}	yotta yotta	Y
10^{21}	zetta zetta	Z
10^{18}	exa exa	E
10^{15}	peta péta	P
10^{12}	tera téra	T
10^{9}	giga giga	G
10^{6}	mega méga	M
10^{3}	kilo kilo	k
10^{2}	hecto hecto	h
10^{1}	deca déca	da
10^{-1}	deci déci	d
10^{-2}	centi centi	c
10^{-3}	milli milli	m
10^{-6}	micro micro	*u*
10^{-9}	nano nano	n

10^{-12}	pico pico	p
10^{-15}	femto femto	f
10^{-18}	atto atto	a
10^{-21}	zepto zepto	z
10^{-24}	yocto yocto	y

1.17.
(1.16) multiple of a unit
measurement unit obtained by multiplying a given measurement unit by an integer greater than one
Prefixes for binary multiples are:

Factor **Facteur**	**Prefix** **Préfixe**	
	Name **Nom**	**Symbol** **Symbole**
$(2^{10})^8$	yobi yobi	Yi
$(2^{10})^7$	zebi zébi	Zi
$(2^{10})^6$	exbi exbi	Ei
$(2^{10})^5$	pebi pébi	Pi
$(2^{10})^4$	tebi tébi	Ti
$(2^{10})^3$	gibi gibi	Gi
$(2^{10})^2$	mebi mébi	Mi
$(2^{10})^1$	kibi kibi	Ki

Source: [IEC 80000-13].

Note 1 to entry: SI prefixes for decimal multiples of SI **base units** and SI **derived units** are given in 1.16, Note 5.

Note 2 to entry: SI prefixes refer strictly to powers of 10, and should not be used for powers of 2. For example, 1 kilobit should not be used to represent 1 024 bits (2^{10} bits), which is 1 kibibit.

EXAMPLE 1 The kilometre is a decimal multiple of the metre.

EXAMPLE 2 The hour is a non-decimal multiple of the second.

1.18.
(1.17) submultiple of a unit
measurement unit obtained by dividing a given measurement unit by an integer greater than one

Note 1 to entry: SI prefixes for decimal submultiples of SI **base units** and SI **derived units** are given in 1.16, Note 5.

EXAMPLE 1 The millimetre is a decimal submultiple of the metre.

EXAMPLE 2 For a plane angle, the second is a non-decimal submultiple of the minute.

1.19.
(1.18) quantity value
value of a quantity
value

number and reference together expressing magnitude of a **quantity**

Note 1 to entry: According to the type of reference, a quantity value is either

- a product of a number and a **measurement unit** (see Examples 1, 2, 3, 4, 5, 8, and 9); the measurement unit one is generally not indicated for **quantities of dimension one** (see Examples 6 and 8), or
- a number and a reference to a **measurement procedure** (see Example 7), or
- a number and a **reference material** (see Example 10).

Note 2 to entry: The number can be complex (see Example 5).

Note 3 to entry: A quantity value can be presented in more than one way (see Examples 1, 2, and 8).

Note 4 to entry: In the case of vector or tensor quantities, each component has a quantity value.

EXAMPLE 1 Length of a given rod: 5.34 m or 534 cm

EXAMPLE 2 Mass of a given body: 0.152 kg or 152 g

EXAMPLE 3 Curvature of a given arc: 112 m^{1}

EXAMPLE 4 Celsius temperature of a given sample: 5 °C

EXAMPLE 5 Electric impedance of a given circuit element at a given frequency, where j is the imaginary unit: (7 3j)

EXAMPLE 6 Refractive index of a given sample of glass: 1.32

EXAMPLE 7 Rockwell C hardness of a given sample: 43.5 HRC

EXAMPLE 8 Mass fraction of cadmium in a given sample of copper: $3\ \mu\mathrm{g} \cdot \mathrm{kg}^{-1}$ or $3 \cdot 10^{-9}$

EXAMPLE 9 Molality of Pb^{2+} in a given sample of water: $1.76\ \mu\mathrm{mol} \cdot \mathrm{kg}^{-1}$

EXAMPLE 10 Arbitrary amount-of-substance concentration of lutropin in a given sample of human blood plasma (WHO International Standard 80/552 used as a calibrator): 5.0 IU/l, where “IU” stands for “WHO International Unit”

EXAMPLE 11 Force acting on a given particle, e.g. in Cartesian components $(F_x; F_y; F_z) = (31.5; 43.2; 17.0)$ N.

1.20.
(1.21) numerical quantity value
numerical value of a quantity numerical value

number in the expression of a **quantity value**, other than any number serving as the reference

Note 1 to entry: For **quantities of dimension one**, the reference is a **measurement unit** which is a number and this is not considered as a part of the numerical quantity value.

Note 2 to entry: For **quantities** that have a measurement unit (i.e. those other than **ordinal quantities**), the numerical value $\{Q\}$ of a quantity Q is frequently denoted $\{Q\} = Q/[Q]$, where $[Q]$ denotes the measurement unit.

EXAMPLE 1 In an amount-of-substance fraction equal to 3 mmol · mol^{-1}, the numerical quantity value is 3 and the unit is mmol · mol^{-1}. The unit mmol · mol^{-1} is numerically equal to 0.001, but this number 0.001 is not part of the numerical quantity value, which remains 3.

EXAMPLE 2 For a quantity value of 5.7 kg, the numerical quantity value is $\{m\} = (5.7\ \text{kg}) / \text{kg} = 5.7$. The same quantity value can be expressed as 5700 g in which case the numerical quantity value $\{m\} = (5700\ \text{g}) / \text{g} = 5700$.

1.21.
quantity calculus
set of mathematical rules and operations applied to **quantities** other than **ordinal quantities**

Note 1 to entry: In quantity calculus, **quantity equations** are preferred to **numerical value equations** because quantity equations are independent of the choice of **measurement units**, whereas numerical value equations are not (see [ISO 31-0:1992], 2.2.2).

1.22.
quantity equation
mathematical relation between **quantities** in a given **system of quantities**, independent of **measurement units**

EXAMPLE 1 $Q_1 = Q_2Q_3$ where Q_1, Q_2 and Q_3 denote different quantities, and where is a numerical factor.

EXAMPLE 2 $T = (1/2)mv^2$ where T is the kinetic energy and v the speed of a specified particle of mass m.

EXAMPLE 3 $n = It/F$ where n is the amount of substance of a univalent component, I is the electric current and t the duration of the electrolysis, and where F is the Faraday constant.

1.23.
unit equation
mathematical relation between **base units**, **coherent derived units** or other **measurement units**

EXAMPLE 1 For the **quantities** in 1.22, Example 1, $[Q_1] = [Q_2][Q_3]$ where, $[Q_1]$, $[Q_2]$, and $[Q_3]$ denote the measurement units of Q_1, Q_2, and Q_3, respectively, provided that these measurement units are in a **coherent system of units** .

EXAMPLE 2 $\text{J} := \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$, where, J, kg, m, and s are the symbols for the joule, kilogram, metre, and second, respectively. (The symbol := denotes “is by definition equal to” as given in the ISO 80000 and IEC 80000 series.)

EXAMPLE 3 $1\ \text{km} \cdot \text{h}^{-1} = \left(\frac{1}{3.6}\right) \text{m} \cdot \text{s}^{-1}$.

1.24.
conversion factor between units
ratio of two **measurement units** for **quantities** of the same **kind**

Note 1 to entry: The measurement units may belong to different **systems of units**.

EXAMPLE 1 $\text{km} \cdot \text{m}^{-1} = 1000$ and thus $1\ \text{km} = 1000\ \text{m}$.

EXAMPLE 2 $\text{h} \cdot \text{s}^{-1} = 3600$ and thus $1\ \text{h} = 3600\ \text{s}$.

EXAMPLE 3 $\frac{\text{unistml}(\text{m/s})}{\text{km}\cdot\text{h}^{-1}} = (1/3.6)$ and thus $1\ \text{km} \cdot \text{h}^{-1} = (1/3.6)\ \text{m} \cdot \text{s}^{-1}$.

1.25.
numerical value equation
numerical quantity value equation

mathematical relation between **numerical quantity values**, based on a given **quantity equation** and specified **measurement units**

EXAMPLE 1 For the **quantities** in 1.22, Example 1, $\{Q_1\} = \{Q_2\}\{Q_3\}$, where $\{Q_1\}$, $\{Q_2\}$, and $\{Q_3\}$ denote the numerical values of Q_1, Q_2, and Q_3, respectively, provided that they are expressed in either **base units** or **coherent derived units** or both.

EXAMPLE 2 In the quantity equation for kinetic energy of a particle, $T = (1/2)mv^2$, if $m = 2$ kg and $v = 3 \text{ m} \cdot \text{s}^{-1}$, then $\{T\} = (1/2)\ 2\ 3^2$ is a numerical value equation giving the numerical value 9 of T in joules.

1.26.
ordinal quantity
quantity, defined by a conventional **measurement procedure**, for which a total ordering relation can be established, according to magnitude, with other quantities of the same **kind**, but for which no algebraic operations among those quantities exist

Note 1 to entry: Ordinal quantities can enter into empirical relations only and have neither **measurement units** nor **quantity dimensions**. Differences and ratios of ordinal quantities have no physical meaning.

Note 2 to entry: Ordinal quantities are arranged according to **ordinal quantity-value scales** (see 1.28).

EXAMPLE 1 Rockwell C hardness.

EXAMPLE 2 Octane number for petroleum fuel.

EXAMPLE 3 Earthquake strength on the Richter scale.

EXAMPLE 4 Subjective level of abdominal pain on a scale from zero to five.

1.27.
quantity-value scale
measurement scale

ordered set of **quantity values** of **quantities** of a given **kind of quantity** used in ranking, according to magnitude, quantities of that kind

EXAMPLE 1 Celsius temperature scale.

EXAMPLE 2 Time scale.

EXAMPLE 3 Rockwell C hardness scale.

1.28.
(1.22) ordinal quantity-value scale
ordinal value scale

quantity-value scale for **ordinal quantities**

Note 1 to entry: An ordinal quantity-value scale may be established by **measurements** according to a **measurement procedure**.

EXAMPLE 1 Rockwell C hardness scale.

EXAMPLE 2 Scale of octane numbers for petroleum fuel.

1.29.
conventional reference scale
quantity-value scale defined by formal agreement

1.30.
nominal property
property of a phenomenon, body, or substance, where the property has no magnitude

Note 1 to entry: A nominal property has a value, which can be expressed in words, by alphanumerical codes, or by other means.

Note 2 to entry: ‘Nominal property value’ is not to be confused with **nominal quantity value**.

EXAMPLE 1 Sex of a human being.

EXAMPLE 2 Colour of a paint sample.

EXAMPLE 3 Colour of a spot test in chemistry.

EXAMPLE 4 ISO two-letter country code.

EXAMPLE 5 Sequence of amino acids in a polypeptide.

2. Measurement

For the purposes of this document, the following terms and definitions apply.

2.1.
(2.1) measurement
process of experimentally obtaining one or more **quantity values** that can reasonably be attributed to a **quantity**

Note 1 to entry: Measurement does not apply to **nominal properties**.

Note 2 to entry: Measurement implies comparison of quantities or counting of entities.

Note 3 to entry: Measurement presupposes a description of the quantity commensurate with the intended use of a **measurement result**, a **measurement procedure**, and a calibrated **measuring system** operating according to the specified measurement procedure, including the measurement conditions.

2.2.
(2.2) metrology
science of **measurement** and its application

Note 1 to entry: Metrology includes all theoretical and practical aspects of measurement, whatever the **measurement uncertainty** and field of application.

2.3.
(2.6) measurand
quantity intended to be measured

Note 1 to entry: The specification of a measurand requires knowledge of the **kind of quantity**, description of the state of the phenomenon, body, or substance carrying the quantity, including any relevant component, and the chemical entities involved.

Note 2 to entry: In the second edition of the VIM and in [IEC 60050-300:2001], the measurand is defined as the “particular quantity subject to measurement”.

Note 3 to entry: The **measurement**, including the **measuring system** and the conditions under which the measurement is carried out, might change the phenomenon, body, or substance such that the quantity being measured may differ from the measurand as defined. In this case, adequate **correction** is necessary.

Note 4 to entry: In chemistry, “analyte”, or the name of a substance or compound, are terms sometimes used for ‘measurand’. This usage is erroneous because these terms do not refer to quantities.

EXAMPLE 1 The potential difference between the terminals of a battery may decrease when using a voltmeter with a significant internal conductance to perform the measurement. The open-circuit potential difference can be calculated from the internal resistances of the battery and the voltmeter.

EXAMPLE 2 The length of a steel rod in equilibrium with the ambient Celsius temperature of 23 °C will be different from the length at the specified temperature of 20 °C, which is the measurand. In this case, a correction is necessary.

2.4.
(2.3) measurement principle
principle of measurement

phenomenon serving as a basis of a **measurement**

Note 1 to entry: The phenomenon can be of a physical, chemical, or biological nature.

EXAMPLE 1 Thermoelectric effect applied to the measurement of temperature.

EXAMPLE 2 Energy absorption applied to the measurement of amount-of-substance concentration.

EXAMPLE 3 Lowering of the concentration of glucose in blood in a fasting rabbit applied to the measurement of insulin concentration in a preparation.

2.5.
(2.4) measurement method
method of measurement

generic description of a logical organization of operations used in a **measurement**

Note 1 to entry: Measurement methods may be qualified in various ways such as:

— substitution measurement method,

— differential measurement method, and

— null measurement method;

or

— direct measurement method, and

— indirect measurement method.

See [IEC 60050-300:2001].

2.6.
(2.5) measurement procedure
detailed description of a **measurement** according to one or more **measurement principles** and to a given **measurement method**, based on a **measurement model** and including any calculation to obtain a **measurement result**

Note 1 to entry: A measurement procedure is usually documented in sufficient detail to enable an operator to perform a measurement.

Note 2 to entry: A measurement procedure can include a statement concerning a **target measurement uncertainty**.

Note 3 to entry: A measurement procedure is sometimes called a standard operating procedure, abbreviated SOP.

2.7.
reference measurement procedure
measurement procedure accepted as providing **measurement results** fit for their intended use in assessing **measurement trueness** of **measured quantity values** obtained from other measurement procedures for **quantities** of the same **kind**, in **calibration**, or in characterizing **reference materials**

2.8.
primary reference measurement procedure
primary reference procedure

reference measurement procedure used to obtain a **measurement result** without relation to a **measurement standard** for a **quantity** of the same **kind**

Note 1 to entry: The Consultative Committee for Amount of Substance — Metrology in Chemistry (CCQM) uses the term "primary method of measurement" for this concept.

Note 2 to entry: Definitions of two subordinate concepts, which could be termed "direct primary reference measurement procedure" and "ratio primary reference measurement procedure", are given by the CCQM ([CCQM Meeting 5]).

EXAMPLE The volume of water delivered by a 50 ml pipette at 20 °C is measured by weighing the water delivered by the pipette into a beaker, taking the mass of beaker plus water minus the mass of the initially empty beaker, and correcting the mass difference for the actual water temperature using the volumic mass (mass density).

2.9.
(3.1) measurement result
result of measurement

set of **quantity values** being attributed to a **measurand** together with any other available relevant information

Note 1 to entry: A measurement result generally contains "relevant information" about the set of quantity values, such that some may be more representative of the measurand than others. This may be expressed in the form of a probability density function (PDF).

Note 2 to entry: A measurement result is generally expressed as a single **measured quantity value** and a **measurement uncertainty**. If the measurement uncertainty is considered to be negligible for some purpose, the measurement result may be expressed as a single measured quantity value. In many fields, this is the common way of expressing a measurement result.

Note 3 to entry: In the traditional literature and in the previous edition of the VIM, measurement result was defined as a value attributed to a measurand and explained to mean an **indication**, or an uncorrected result, or a corrected result, according to the context.

2.10.
measured quantity value
value of a measured
quantity measured value

quantity value representing a **measurement result**

Note 1 to entry: For a **measurement** involving replicate **indications**, each indication can be used to provide a corresponding measured quantity value. This set of individual measured quantity values can be used to calculate a resulting measured quantity value, such as an average or median, usually with a decreased associated **measurement uncertainty**.

Note 2 to entry: When the range of the **true quantity values** believed to represent the **measurand** is small compared with the measurement uncertainty, a measured quantity value can be considered to be an estimate of an essentially unique true quantity value and is often an average or median of individual measured quantity values obtained through replicate measurements.

Note 3 to entry: In the case where the range of the true quantity values believed to represent the measurand is not small compared with the measurement uncertainty, a measured quantity value is often an estimate of an average or median of the set of true quantity values.

Note 4 to entry: In the GUM, the terms "result of measurement" and "estimate of the value of the measurand" or just "estimate of the measurand" are used for 'measured quantity value'.

2.11.
(1.19) true quantity value
true value of a quantity
true value

quantity value consistent with the definition of a **quantity**

Note 1 to entry: In the Error Approach to describing **measurement**, a true quantity value is considered unique and, in practice, unknowable. The Uncertainty Approach is to recognize that, owing to the inherently incomplete amount of detail in the definition of a quantity, there is not a single true quantity value but rather a set of true quantity values consistent with the definition. However, this set of values is, in principle and in practice, unknowable. Other approaches dispense altogether with the concept of true quantity value and rely on the concept of **metrological compatibility of measurement results** for assessing their validity.

Note 2 to entry: In the special case of a fundamental constant, the quantity is considered to have a single true quantity value.

Note 3 to entry: When the **definitional uncertainty** associated with the **measurand** is considered to be negligible compared to the other components of the **measurement uncertainty**, the measurand may be considered to have an "essentially unique" true quantity value. This is the approach taken by the GUM and associated documents, where the word "true" is considered to be redundant.

2.12.
conventional quantity value
conventional value of a quantity
conventional value

quantity value attributed by agreement to a **quantity** for a given purpose

Note 1 to entry: The term "conventional true quantity value" is sometimes used for this concept, but its use is discouraged.

Note 2 to entry: Sometimes a conventional quantity value is an estimate of a **true quantity value**.

Note 3 to entry: A conventional quantity value is generally accepted as being associated with a suitably small **measurement uncertainty**, which might be zero.

EXAMPLE 1 Standard acceleration of free fall (formerly called "standard acceleration due to gravity"), $g_n = 9.80665\ \mathrm{m} \cdot \mathrm{s}^2$.

EXAMPLE 2 Conventional quantity value of the Josephson constant, $K_{J\text{-}90} = 483597.9\ \mathrm{GHz} \cdot \mathrm{V}^1$

EXAMPLE 3 Conventional quantity value of a given mass standard, $m = 100.003\ 47$ g.

2.13.
(3.5) measurement accuracy
accuracy of measurement
accuracy

closeness of agreement between a **measured quantity value** and a **true quantity value** of a **measurand**

Note 1 to entry: The concept 'measurement accuracy' is not a **quantity** and is not given a **numerical quantity value**. A **measurement** is said to be more accurate when it offers a smaller **measurement error**.

Note 2 to entry: The term "measurement accuracy" should not be used for **measurement trueness** and the term "measurement precision" should not be used for 'measurement accuracy', which, however, is related to both these concepts.

Note 3 to entry: 'Measurement accuracy' is sometimes understood as closeness of agreement between measured quantity values that are being attributed to the measurand.

2.14.
measurement trueness
trueness of measurement
trueness

closeness of agreement between the average of an infinite number of replicate **measured quantity values** and a **reference quantity value**

Note 1 to entry: Measurement trueness is not a **quantity** and thus cannot be expressed numerically, but measures for closeness of agreement are given in ISO 5725.

Note 2 to entry: Measurement trueness is inversely related to **systematic measurement error**, but is not related to **random measurement error**.

Note 3 to entry: “Measurement accuracy” should not be used for ‘measurement trueness’.

2.15.
measurement precision
precision

closeness of agreement between **indications** or **measured quantity values** obtained by replicate **measurements** on the same or similar objects under specified conditions

Note 1 to entry: Measurement precision is usually expressed numerically by measures of imprecision, such as standard deviation, variance, or coefficient of variation under the specified conditions of measurement.

Note 2 to entry: The ‘specified conditions’ can be, for example, **repeatability conditions of measurement**, **intermediate precision conditions of measurement** , or **reproducibility conditions of measurement** (see [ISO 5725-1:1994/Cor 1:1998]).

Note 3 to entry: Measurement precision is used to define **measurement repeatability**, **intermediate measurement precision**, and **measurement reproducibility**.

Note 4 to entry: Sometimes “measurement precision” is erroneously used to mean **measurement accuracy**.

2.16.
(3.10) measurement error
error of measurement
error

measured quantity value minus a **reference quantity value**

Note 1 to entry: The concept of ‘measurement error’ can be used both

1. when there is a single reference quantity value to refer to, which occurs if a **calibration** is made by means of a **measurement standard** with a **measured quantity value** having a negligible **measurement uncertainty** or if a **conventional quantity value** is given, in which case the measurement error is known, and

2. if a **measurand** is supposed to be represented by a unique **true quantity value** or a set of true quantity values of negligible range, in which case the measurement error is not known.

Note 2 to entry: Measurement error should not be confused with production error or mistake.

2.17.
(3.14) systematic measurement
error systematic error of measurement
systematic error

component of **measurement error** that in replicate **measurements** remains constant or varies in a predictable manner

Note 1 to entry: A **reference quantity value** for a systematic measurement error is a **true quantity value**, or a **measured quantity value** of a **measurement standard** of negligible **measurement uncertainty**, or a **conventional quantity value**.

Note 2 to entry: Systematic measurement error, and its causes, can be known or unknown. A **correction** can be applied to compensate for a known systematic measurement error.

Note 3 to entry: Systematic measurement error equals measurement error minus **random measurement error**.

2.18.
measurement bias
bias

estimate of a **systematic measurement error**

2.19.
(3.13) random measurement error
random error of measurement
random error

component of **measurement error** that in replicate **measurements** varies in an unpredictable manner

Note 1 to entry: A **reference quantity value** for a random measurement error is the average that would ensue from an infinite number of replicate measurements of the same **measurand**.

Note 2 to entry: Random measurement errors of a set of replicate measurements form a distribution that can be summarized by its expectation, which is generally assumed to be zero, and its variance.

Note 3 to entry: Random measurement error equals measurement error minus **systematic measurement error**.

2.20.
(3.6, Notes 1 and 2) repeatability condition of measurement
repeatability condition

condition of **measurement** , out of a set of conditions that includes the same **measurement procedure**, same operators, same **measuring system**, same operating conditions and same location, and replicate measurements on the same or similar objects over a short period of time

Note 1 to entry: A condition of measurement is a repeatability condition only with respect to a specified set of repeatability conditions.

Note 2 to entry: In chemistry, the term “intra-serial precision condition of measurement” is sometimes used to designate this concept.

2.21.
(3.6) measurement repeatability
repeatability

measurement precision under a set of **repeatability conditions of measurement**

2.22.
intermediate precision condition of measurement
intermediate precision condition

condition of **measurement** , out of a set of conditions that includes the same **measurement procedure**, same location, and replicate measurements on the same or similar objects over an extended period of time, but may include other conditions involving changes

Note 1 to entry: The changes can include new **calibrations**, **calibrators**, operators, and **measuring systems**.

Note 2 to entry: A specification for the conditions should contain the conditions changed and unchanged, to the extent practical.

Note 3 to entry: In chemistry, the term “inter-serial precision condition of measurement” is sometimes used to designate this concept.

2.23.
intermediate measurement precision
intermediate precision

measurement precision under a set of **intermediate precision conditions of measurement**

Note 1 to entry: Relevant statistical terms are given in [ISO 5725-3:1994/Cor 1:2001].

2.24.
(3.7, Note 2) reproducibility condition of measurement
reproducibility condition

condition of **measurement**, out of a set of conditions that includes different locations, operators, **measuring systems**, and replicate measurements on the same or similar objects

Note 1 to entry: The different measuring systems may use different **measurement procedures**.

Note 2 to entry: A specification should give the conditions changed and unchanged, to the extent practical.

2.25.
(3.7) measurement reproducibility
reproducibility

measurement precision under **reproducibility conditions of measurement**

Note 1 to entry: Relevant statistical terms are given in [ISO 5725-1:1994/Cor 1:1998] and [ISO 5725-2:1994/Cor 1:2002].

2.26.
(3.9) measurement uncertainty
uncertainty of measurement uncertainty

non-negative parameter characterizing the dispersion of the **quantity values** being attributed to a **measurand**, based on the information used

Note 1 to entry: Measurement uncertainty includes components arising from systematic effects, such as components associated with **corrections** and the assigned quantity values of **measurement standards**, as well as the **definitional uncertainty**. Sometimes estimated systematic effects are not corrected for but, instead, associated measurement uncertainty components are incorporated.

Note 2 to entry: The parameter may be, for example, a standard deviation called **standard measurement uncertainty** (or a specified multiple of it), or the half-width of an interval, having a stated **coverage probability**.

Note 3 to entry: Measurement uncertainty comprises, in general, many components. Some of these may be evaluated by **Type A evaluation of measurement uncertainty** from the statistical distribution of the quantity values from series of **measurements** and can be characterized by standard deviations. The other components, which may be evaluated by **Type B evaluation of measurement uncertainty** , can also be characterized by standard deviations, evaluated from probability density functions based on experience or other information.

Note 4 to entry: In general, for a given set of information, it is understood that the measurement uncertainty is associated with a stated quantity value attributed to the measurand. A modification of this value results in a modification of the associated uncertainty.

2.27.
definitional uncertainty
component of **measurement uncertainty** resulting from the finite amount of detail in the definition of a **measurand**

Note 1 to entry: Definitional uncertainty is the practical minimum measurement uncertainty achievable in any **measurement** of a given measurand.

Note 2 to entry: Any change in the descriptive detail leads to another definitional uncertainty.

Note 3 to entry: In the [Guide 1995], D.3.4, and in IEC 60359, the concept 'definitional uncertainty' is termed "intrinsic uncertainty".

2.28.
Type A evaluation of measurement uncertainty
Type A evaluation

evaluation of a component of **measurement uncertainty** by a statistical analysis of **measured quantity values** obtained under defined measurement conditions

Note 1 to entry: For various types of measurement conditions, see **repeatability condition of measurement**, **intermediate precision condition of measurement**, and **reproducibility condition of measurement**.

Note 2 to entry: For information about statistical analysis, see e.g.the [Guide 1995].

Note 3 to entry: See also [Guide 1995], 2.3.2, ISO 5725, [ISO 13528, ISO/TS 21748, ISO/TS 21749:2005].

2.29.
Type B evaluation of measurement uncertainty
Type B evaluation

evaluation of a component of **measurement uncertainty** determined by means other than a **Type A evaluation of measurement uncertainty**

Note 1 to entry: See also [Guide 1995], 2.3.3.

EXAMPLE Evaluation based on information

— associated with authoritative published **quantity values**,

— associated with the quantity value of a **certified reference material**,

— obtained from a **calibration** certificate,

— about drift,

— obtained from the **accuracy class** of a verified **measuring instrument**,

— obtained from limits deduced through personal experience.

2.30.
standard measurement uncertainty
standard uncertainty of measurement
standard uncertainty

measurement uncertainty expressed as a standard deviation

2.31.
combined standard measurement uncertainty
combined standard uncertainty

standard measurement uncertainty that is obtained using the individual standard measurement uncertainties associated with the **input quantities in a measurement model**

Note 1 to entry: In case of correlations of input quantities in a measurement model, covariances must also be taken into account when calculating the combined standard measurement uncertainty; see also [Guide 1995], 2.3.4.

2.32.
relative standard measurement uncertainty
standard measurement uncertainty divided by the absolute value of the **measured quantity value**

2.33.
uncertainty budget
statement of a **measurement uncertainty**, of the components of that measurement uncertainty, and of their calculation and combination

Note 1 to entry: An uncertainty budget should include the **measurement model**, estimates, and measurement uncertainties associated with the **quantities** in the measurement model, covariances, type of applied probability density functions, degrees of freedom, type of evaluation of measurement uncertainty, and any **coverage factor**.

2.34.
target measurement uncertainty
target uncertainty

measurement uncertainty specified as an upper limit and decided on the basis of the intended use of **measurement results**

2.35.
expanded measurement uncertainty
expanded uncertainty

product of a **combined standard measurement uncertainty** and a factor larger than the number one

Note 1 to entry: The factor depends upon the type of probability distribution of the **output quantity in a measurement model** and on the selected **coverage probability**.

Note 2 to entry: The term “factor” in this definition refers to a **coverage factor**.

Note 3 to entry: Expanded measurement uncertainty is termed “overall uncertainty” in paragraph 5 of Recommendation INC-1 (1980) (see the GUM) and simply “uncertainty” in IEC documents.

2.36.
coverage interval
interval containing the set of **true quantity values** of a **measurand** with a stated probability, based on the information available

Note 1 to entry: A coverage interval does not need to be centred on the chosen **measured quantity value** (see [JCGM 101:2008]).

Note 2 to entry: A coverage interval should not be termed “confidence interval” to avoid confusion with the statistical concept (see [Guide 1995], 6.2.2).

Note 3 to entry: A coverage interval can be derived from an **expanded measurement uncertainty** (see [Guide 1995], 2.3.5).

2.37.
coverage probability
probability that the set of **true quantity values** of a **measurand** is contained within a specified **coverage interval**

Note 1 to entry: This definition pertains to the Uncertainty Approach as presented in the GUM.

Note 2 to entry: The coverage probability is also termed “level of confidence” in the GUM.

2.38.
coverage factor
number larger than one by which a **combined standard measurement uncertainty** is multiplied to obtain an **expanded measurement uncertainty**

Note 1 to entry: A coverage factor is usually symbolized k (see also [Guide 1995], 2.3.6).

2.39.
(6.11) calibration
operation that, under specified conditions, in a first step, establishes a relation between the **quantity values** with **measurement uncertainties** provided by **measurement standards** and corresponding **indications** with associated measurement uncertainties and, in a second step, uses this information to establish a relation for obtaining a **measurement result** from an indication

Note 1 to entry: A calibration may be expressed by a statement, calibration function, **calibration diagram**, **calibration curve**, or calibration table. In some cases, it may consist of an additive or multiplicative **correction** of the indication with associated measurement uncertainty.

Note 2 to entry: Calibration should not be confused with **adjustment of a measuring system**, often mistakenly called "self-calibration", nor with **verification** of calibration.

Note 3 to entry: Often, the first step alone in the above definition is perceived as being calibration.

2.40.
calibration hierarchy
sequence of **calibrations** from a reference to the final **measuring system**, where the outcome of each calibration depends on the outcome of the previous calibration

Note 1 to entry: **Measurement uncertainty** necessarily increases along the sequence of calibrations.

Note 2 to entry: The elements of a calibration hierarchy are one or more **measurement standards** and measuring systems operated according to **measurement procedures**.

Note 3 to entry: For this definition, the 'reference' can be a definition of a **measurement unit** through its practical realization, or a measurement procedure, or a measurement standard.

Note 4 to entry: A comparison between two measurement standards may be viewed as a calibration if the comparison is used to check and, if necessary, correct the **quantity value** and measurement uncertainty attributed to one of the measurement standards.

2.41.
(6.10) metrological traceability
property of a **measurement result** whereby the result can be related to a reference through a documented unbroken chain of **calibrations**, each contributing to the **measurement uncertainty**

Note 1 to entry: For this definition, a 'reference' can be a definition of a **measurement unit** through its practical realization, or a **measurement procedure** including the measurement unit for a non- **ordinal quantity**, or a **measurement standard**.

Note 2 to entry: Metrological traceability requires an established **calibration hierarchy**.

Note 3 to entry: Specification of the reference must include the time at which this reference was used in establishing the calibration hierarchy, along with any other relevant metrological information about the reference, such as when the first calibration in the calibration hierarchy was performed.

Note 4 to entry: For **measurements** with more than one **input quantity in the measurement model** , each of the input **quantity values** should itself be metrologically traceable and the calibration hierarchy involved may form a branched structure or a network. The effort involved in establishing metrological traceability for each input quantity value should be commensurate with its relative contribution to the measurement result.

Note 5 to entry: Metrological traceability of a measurement result does not ensure that the measurement uncertainty is adequate for a given purpose or that there is an absence of mistakes.

Note 6 to entry: A comparison between two measurement standards may be viewed as a calibration if the comparison is used to check and, if necessary, correct the quantity value and measurement uncertainty attributed to one of the measurement standards.

Note 7 to entry: The ILAC considers the elements for confirming metrological traceability to be an unbroken **metrological traceability chain** to an **international measurement standard** or a **national measurement standard** , a documented measurement uncertainty, a documented measurement procedure, accredited technical competence, metrological traceability to the SI, and calibration intervals (see [ILAC P-10:2002]).

Note 8 to entry: The abbreviated term “traceability” is sometimes used to mean ‘metrological traceability’ as well as other concepts, such as ‘sample traceability’ or ‘document traceability’ or ‘instrument traceability’ or ‘material traceability’, where the history (“trace”) of an item is meant. Therefore, the full term of “metrological traceability” is preferred if there is any risk of confusion.

2.42.
(6.10 Note 2) metrological traceability chain
traceability chain

sequence of **measurement standards** and **calibrations** that is used to relate a **measurement result** to a reference

Note 1 to entry: A metrological traceability chain is defined through a **calibration hierarchy**.

Note 2 to entry: A metrological traceability chain is used to establish **metrological traceability** of a measurement result.

Note 3 to entry: A comparison between two measurement standards may be viewed as a calibration if the comparison is used to check and, if necessary, correct the **quantity value** and **measurement uncertainty** attributed to one of the measurement standards.

2.43.
metrological traceability to a measurement unit
metrological traceability to a unit

metrological traceability where the reference is the definition of a **measurement unit** through its practical realization

Note 1 to entry: The expression “traceability to the SI” means ‘metrological traceability to a measurement unit of the **International System of Units** ‘.

2.44.
verification
provision of objective evidence that a given item fulfils specified requirements

Note 1 to entry: When applicable, **measurement uncertainty** should be taken into consideration.

Note 2 to entry: The item may be, e.g. a process, measurement procedure, material, compound, or measuring system.

Note 3 to entry: The specified requirements may be, e.g. that a manufacturer’s specifications are met.

Note 4 to entry: Verification in legal metrology, as defined in VIML, and in conformity assessment in general, pertains to the examination and marking and/or issuing of a verification certificate for a measuring system.

Note 5 to entry: Verification should not be confused with **calibration**. Not every verification is a **validation**.

Note 6 to entry: In chemistry, verification of the identity of the entity involved, or of activity, requires a description of the structure or properties of that entity or activity.

EXAMPLE 1 Confirmation that a given **reference material** as claimed is homogeneous for the **quantity value** and **measurement procedure** concerned, down toa measurement portion having a mass of 10 mg.

EXAMPLE 2 Confirmation that performance properties or legal requirements of a **measuring system** are achieved.

EXAMPLE 3 Confirmation that a **target measurement uncertainty** can be met.

2.45.
validation
verification, where the specified requirements are adequate for an intended use

EXAMPLE A **measurement procedure**, ordinarily used for the **measurement** of mass concentration of nitrogen in water, may be validated also for measurement of mass concentration of nitrogen in human serum.

2.46.
metrological comparability of measurement results
metrological comparability

comparability of **measurement results**, for **quantities** of a given **kind**, that are metrologically traceable to the same reference

Note 1 to entry: See 2.41, Note 1 **metrological traceability**.

Note 2 to entry: Metrological comparability of measurement results does not necessitate that the **measured quantity values** and associated **measurement uncertainties** compared be of the same order of magnitude.

EXAMPLE Measurement results, for the distances between the Earth and the Moon, and between Paris and London, are metrologically comparable when they are both metrologically traceable to the same **measurement unit**, for instance the metre.

2.47.
metrological compatibility of measurement results
metrological compatibility

property of a set of **measurement results** for a specified **measurand**, such that the absolute value of the difference of any pair of **measured quantity values** from two different measurement results is smaller than some chosen multiple of the **standard measurement uncertainty** of that difference

Note 1 to entry: Metrological compatibility of measurement results replaces the traditional concept of 'staying within the error', as it represents the criterion for deciding whether two measurement results refer to the same measurand or not. If in a set of **measurements** of a measurand, thought to be constant, a measurement result is not compatible with the others, either the measurement was not correct (e.g. its **measurement uncertainty** was assessed as being too small) or the measured **quantity** changed between measurements.

Note 2 to entry: Correlation between the measurements influences metrological compatibility of measurement results. If the measurements are completely uncorrelated, the standard measurement uncertainty of their difference is equal to the root mean square sum of their standard measurement uncertainties, while it is lower for positive covariance or higher for negative covariance.

2.48.
measurement model
model of measurement
model

mathematical relation among all **quantities** known to be involved in a **measurement**

Note 1 to entry: A general form of a measurement model is the equation $h(\mathrm{Y}, \mathrm{X}_1, \ , \mathrm{X}_n) = 0$, where Y, the **output quantity in the measurement model**, is the **measurand**, the **quantity value** of which is to be inferred from information about **input quantities in the measurement model** $\mathrm{X}_1, ,\mathrm{X}_n$.

Note 2 to entry: In more complex cases where there are two or more output quantities in a measurement model, the measurement model consists of more than one equation.

2.49.
measurement function
function of **quantities**, the value of which, when calculated using known **quantity values** for the **input quantities in a measurement model** , is a **measured quantity value** of the **output quantity in the measurement model**

Note 1 to entry: If a **measurement model** $h(\mathrm{Y}, \mathrm{X}_1, \ , \mathrm{X}_n) = 0$ can explicitly be written as $\mathrm{Y} = f(\mathrm{X}_1, \ , \mathrm{X}_n)$, where Y is the output quantity in the measurement model, the function f is the measurement function. More generally, f may symbolize an algorithm, yielding for input quantity values $x_1, , x_n$ a corresponding unique output quantity value $y = f(x_1, \ , x_n)$.

Note 2 to entry: A measurement function is also used to calculate the **measurement uncertainty** associated with the measured quantity value of Y.

2.50.
input quantity in a measurement model
input quantity

quantity that must be measured, or a quantity, the **value** of which can be otherwise obtained, in order to calculate a **measured quantity value** of a **measurand**

Note 1 to entry: An input quantity in a measurement model is often an output quantity of a **measuring system**.

Note 2 to entry: **Indications**, **corrections** and **influence quantities** can be input quantities in a measurement model.

EXAMPLE When the length of a steel rod at a specified temperature is the measurand, the actual temperature, the length at that actual temperature, and the linear thermal expansion coefficient of the rod are input quantities in a measurement model.

2.51.
output quantity in a measurement model
output quantity

quantity, the **measured value** of which is calculated using the **values** of **input quantities in a measurement model**

2.52.
(2.7) influence quantity
quantity that, in a direct **measurement**, does not affect the quantity that is actually measured, but affects the relation between the **indication** and the **measurement result**

Note 1 to entry: An indirect measurement involves a combination of direct measurements, each of which may be affected by influence quantities.

Note 2 to entry: In the GUM, the concept ‘influence quantity’ is defined as in the second edition of the VIM, covering not only the quantities affecting the **measuring system**, as in the definition above, but also those quantities that affect the quantities actually measured. Also, in the GUM this concept is not restricted to direct measurements.

EXAMPLE 1 Frequency in the direct measurement with an ammeter of the constant amplitude of an alternating current.

EXAMPLE 2 Amount-of-substance concentration of bilirubin in a direct measurement of haemoglobin amount-of-substance concentration in human blood plasma.

EXAMPLE 3 Temperature of a micrometer used for measuring the length of a rod, but not the temperature of the rod itself which can enter into the definition of the **measurand**.

EXAMPLE 4 Background pressure in the ion source of a mass spectrometer during a measurement of amount-of-substance fraction.

2.53.
(3.15) (3.16) correction
compensation for an estimated systematic effect

Note 1 to entry: See [Guide 1995], 3.2.3, for an explanation of 'systematic effect'.

Note 2 to entry: The compensation can take different forms, such as an addend or a factor, or can be deduced from a table.

3. Devices for measurement

For the purposes of this document, the following terms and definitions apply.

3.1.
(4.1) measuring instrument
device used for making **measurements**, alone or in conjunction with one or more supplementary devices

Note 1 to entry: A measuring instrument that can be used alone is a **measuring system**.

Note 2 to entry: A measuring instrument may be an **indicating measuring instrument** or a **material measure**.

3.2.
(4.5) measuring system
set of one or more **measuring instruments** and often other devices, including any reagent and supply, assembled and adapted to give information used to generate **measured quantity values** within specified intervals for **quantities** of specified **kinds**

Note 1 to entry: A measuring system may consist of only one measuring instrument.

3.3.
(4.6) indicating measuring instrument
measuring instrument providing an output signal carrying information about the **value** of the **quantity** being measured

Note 1 to entry: An indicating measuring instrument may provide a record of its **indication**.

Note 2 to entry: An output signal may be presented in visual or acoustic form. It may also be transmitted to one or more other devices.

EXAMPLE Voltmeter, micrometer, thermometer, electronic balance.

3.4.
(4.6) displaying measuring instrument
indicating measuring instrument where the output signal is presented in visual form

3.5.
(4.17) scale of a displaying measuring instrument
part of a **displaying measuring instrument**, consisting of an ordered set of marks together with any associated **quantity values**

3.6.
(4.2) material measure
measuring instrument reproducing or supplying, in a permanent manner during its use, **quantities** of one or more given **kinds**, each with an assigned **quantity value**

Note 1 to entry: The **indication** of a material measure is its assigned quantity value.

Note 2 to entry: A material measure can be a **measurement standard**.

EXAMPLE Standard weight, volume measure (supplying one or several quantity values, with or without a **quantity-value scale**), standard electric resistor, line scale (ruler), gauge block, standard signal generator, **certified reference material**.

3.7.
(4.3) measuring transducer
device, used in **measurement**, that provides an output **quantity** having a specified relation to the input quantity

EXAMPLE Thermocouple, electric current transformer, strain gauge, pH electrode, Bourdon tube, bimetallic strip.

3.8.
(4.14) sensor
element of a **measuring system** that is directly affected by a phenomenon, body, or substance carrying a **quantity** to be measured

Note 1 to entry: In some fields, the term “detector” is used for this concept.

EXAMPLE Sensing coil of a platinum resistance thermometer, rotor of a turbine flow meter, Bourdon tube of a pressure gauge, float of a level-measuring instrument, photocell of a spectrometer, thermotropic liquid crystal which changes colour as a function of temperature.

3.9.
(4.15) detector
device or substance that indicates the presence of a phenomenon, body, or substance when a threshold **value** of an associated **quantity** is exceeded

Note 1 to entry: In some fields, the term “detector” is used for the concept of **sensor**.

Note 2 to entry: In chemistry, the term “indicator” is frequently used for this concept.

EXAMPLE Halogen leak detector, litmus paper.

3.10.
(4.4) measuring chain
series of elements of a **measuring system** constituting a single path of the signal from a **sensor** to an output element

EXAMPLE 1 Electro-acoustic measuring chain comprising a microphone, attenuator, filter, amplifier, and voltmeter.

EXAMPLE 2 Mechanical measuring chain comprising a Bourdon tube, system of levers, two gears, and a mechanical dial.

3.11.
(4.30) adjustment of a measuring system
adjustment

set of operations carried out on a **measuring system** so that it provides prescribed **indications** corresponding to given **values** of a **quantity** to be measured

Note 1 to entry: Types of adjustment of a measuring system include **zero adjustment of a measuring system**, offset adjustment, and span adjustment (sometimes called gain adjustment).

Note 2 to entry: Adjustment of a measuring system should not be confused with **calibration**, which is a prerequisite for adjustment.

Note 3 to entry: After an adjustment of a measuring system, the measuring system must usually be re-calibrated.

3.12.
zero adjustment of a measuring system
zero adjustment

adjustment of a measuring system so that it provides a null **indication** corresponding to a zero **value** of a **quantity** to be measured

4. Properties of measuring devices

For the purposes of this document, the following terms and definitions apply.

4.1.
(3.2) indication
quantity value provided by a **measuring instrument** or a **measuring system**

Note 1 to entry: An indication may be presented in visual or acoustic form or may be transferred to another device. An indication is often given by the position of a pointer on the display for analog outputs, a displayed or printed number for digital outputs, a code pattern for code outputs, or an assigned quantity value for **material measures**.

Note 2 to entry: An indication and a corresponding value of the **quantity** being measured are not necessarily values of quantities of the same **kind**.

4.2.
blank indication
background indication

indication obtained from a phenomenon, body, or substance similar to the one under investigation, but for which a **quantity** of interest is supposed not to be present, or is not contributing to the indication

4.3.
(4.19) indication interval
set of **quantity values** bounded by extreme possible **indications**

Note 1 to entry: An indication interval is usually stated in terms of its smallest and greatest quantity values, for example "99 V to 201 V".

Note 2 to entry: In some fields, the term is "range of indications".

4.4.
(5.1) nominal indication interval
nominal interval

set of **quantity values**, bounded by rounded or approximate extreme **indications**, obtainable with a particular setting of the controls of a **measuring instrument** or **measuring system** and used to designate that setting

Note 1 to entry: A nominal indication interval is usually stated as its smallest and greatest quantity values, for example "100 V to 200 V".

Note 2 to entry: In some fields, the term is "nominal range".

4.5.
(5.2) range of a nominal indication interval
absolute value of the difference between the extreme **quantity values** of a **nominal indication interval**

Note 1 to entry: Range of a nominal indication interval is sometimes termed "span of a nominal interval".

EXAMPLE For a nominal indication interval of -10 V to +10 V, the range of the nominal indication interval is 20 V.

4.6.
(5.3) nominal quantity value
nominal value

rounded or approximate **value** of a characterizing **quantity** of a **measuring instrument** or **measuring system** that provides guidance for its appropriate use

Note 1 to entry: "Nominal quantity value" and "nominal value" should not be used for "nominal property value".

EXAMPLE 1 100 as the nominal quantity value marked on a standard resistor.

EXAMPLE 2 1 000 ml as the nominal quantity value marked on a single-mark volumetric flask.

EXAMPLE 3 0. lunistml(mol/l) as the nominal quantity value for amount-of-substance concentration of a solution of hydrogen chloride, HCl.

EXAMPLE 4 − 20 °C as a maximum Celsius temperature for storage.

4.7.
(5.4) measuring interval
working interval

set of **values** of **quantities** of the same **kind** that can be measured by a given **measuring instrument** or **measuring system** with specified **instrumental measurement uncertainty**, underdefined conditions

Note 1 to entry: In some fields, the term is "measuring range" or "measurement range".

Note 2 to entry: The lower limit of a measuring interval should not be confused with **detection limit**.

4.8.
steady-state operating condition
operating condition of a **measuring instrument** or **measuring system** in which the relation established by **calibration** remains valid even for a **measurand** varying with time

4.9.
(5.5) rated operating condition
operating condition that must be fulfilled during **measurement** in order that a **measuring instrument** or **measuring system** perform as designed

Note 1 to entry: Rated operating conditions generally specify intervals of **values** for a **quantity** being measured and for any **influence quantity**.

4.10.
(5.6) limiting operating condition
extreme operating condition that a **measuring instrument** or **measuring system** is required to withstand without damage, and without degradation of specified metrological properties, when it is subsequently operated under its **rated operating conditions**

Note 1 to entry: Limiting conditions for storage, transport or operation can differ.

Note 2 to entry: Limiting conditions can include limiting **values** of a **quantity** being measured and of any **influence quantity**.

4.11.
(5.7) reference operating condition
reference condition

operating condition prescribed for evaluating the performance of a **measuring instrument** or **measuring system** or for comparison of **measurement results**

Note 1 to entry: Reference operating conditions specify intervals of **values** of the **measurand** and of the **influence quantities**.

Note 2 to entry: In [IEC 60050-300:2001], item 311-06-02, the term "reference condition" refers to an operating condition under which the specified **instrumental measurement uncertainty** is the smallest possible.

4.12.
(5.10) sensitivity of a measuring system
sensitivity

quotient of the change in an **indication** of a **measuring system** and the corresponding change in a **value** of a

quantity being measured

Note 1 to entry: Sensitivity of a measuring system can depend on the value of the quantity being measured.

Note 2 to entry: The change considered in a value of a quantity being measured must be large compared with the **resolution**.

4.13.
selectivity of a measuring system
selectivity

property of a **measuring system**, used with a specified **measurement procedure**, whereby it provides measured **quantity values** for one or more **measurands** such that the values of each measurand are independent of other measurands or other **quantities** in the phenomenon, body, or substance being investigated

Note 1 to entry: In physics, there is often only one measurand; the other quantities are of the same **kind** as the measurand, and they are input quantities to the measuring system.

Note 2 to entry: In chemistry, the measured quantities often involve different components in the system undergoing measurement and these quantities are not necessarily of the same kind.

Note 3 to entry: In chemistry, selectivity of a measuring system is usually obtained for quantities with selected components in concentrations within stated intervals.

Note 4 to entry: Selectivity as used in physics (see Note 1) is a concept close to specificity as it is sometimes used in chemistry.

EXAMPLE 1 Capability of a measuring system including a mass spectrometer to measure the ion current ratio generated by two specified compounds without disturbance by other specified sources of electric current.

EXAMPLE 2 Capability of a measuring system to measure the power of a signal component at a given frequency without being disturbed by signal components or other signals at other frequencies.

EXAMPLE 3 Capability of a receiver to discriminate between a wanted signal and unwanted signals, often having frequencies slightly different from the frequency of the wanted signal.

EXAMPLE 4 Capability of a measuring system for ionizing radiation to respond to a given radiation to be measured in the presence of concomitant radiation.

EXAMPLE 5 Capability of a measuring system to measure the amount-of-substance concentration of creatininium in blood plasma by a Jaffé procedure without being influenced by the glucose, urate, ketone, and protein concentrations.

EXAMPLE 6 Capability of a mass spectrometer to measure the amount-of-substance abundance of the ^{28}Si isotope and of the ^{30}Si isotope in silicon from a geological deposit without influence between the two, or from the ^{29}Si isotope.

4.14.
resolution
smallest change in a **quantity** being measured that causes a perceptible change in the corresponding **indication**

Note 1 to entry: Resolution can depend on, for example, noise (internal or external) or friction. It may also depend on the **value** of a quantity being measured.

4.15.
(5.12) resolution of a displaying device
smallest difference between displayed **indications** that can be meaningfully distinguished

4.16.
(5.11) discrimination threshold
largest change in a **value** of a **quantity** being measured that causes no detectable change in the corresponding **indication**

Note 1 to entry: Discrimination threshold may depend on, e.g. noise (internal or external) or friction. It can also depend on the value of the quantity being measured and how the change is applied.

4.17.
(5.13) dead band
maximum interval through which a **value** of a **quantity** being measured can be changed in both directions without producing a detectable change in the corresponding **indication**

Note 1 to entry: Dead band can depend on the rate of change.

4.18.
(4.15 Note 1) detection limit
limit of detection

measured quantity value, obtained by a given **measurement procedure**, for which the probability of falsely claiming the absence of a component in a material is , given a probability of falsely claiming its presence

Note 1 to entry: IUPAC recommends default values for and equal to 0.05.

Note 2 to entry: The abbreviation LOD is sometimes used.

Note 3 to entry: The term “sensitivity” is discouraged for ‘detection limit’.

4.19.
(5.14) stability of a measuring instrument
stability

property of a **measuring instrument**, whereby its metrological properties remain constant in time

Note 1 to entry: Stability may be quantified in several ways.

EXAMPLE 1 In terms of the duration of a time interval over which a metrological property changes by a stated amount.

EXAMPLE 2 In terms of the change of a property over a stated time interval.

4.20.
(5.25) instrumental bias
average of replicate **indications** minus a **reference quantity value**

4.21.
(5.16) instrumental drift
continuous or incremental change over time in **indication**, due to changes in metrological properties of a **measuring instrument**

Note 1 to entry: Instrumental drift is related neither to a change in a **quantity** being measured nor to a change of any recognized **influence quantity**.

4.22.
variation due to an influence quantity
difference in **indication** for a given **measured quantity value**, or in **quantity values** supplied by a **material measure**, when an **influence quantity** assumes successively two different quantity values

4.23.
(5.17) step response time
duration between the instant when an input **quantity value** of a **measuring instrument** or **measuring system** is subjected to an abrupt change between two specified constant quantity values and the instant when a corresponding **indication** settles within specified limits around its final steady value

4.24.
instrumental measurement uncertainty
component of **measurement uncertainty** arising from a **measuring instrument** or **measuring system** in use

Note 1 to entry: Instrumental measurement uncertainty is obtained through **calibration** of a measuring instrument or measuring system, except for a **primary measurement standard** for which other means are used.

Note 2 to entry: Instrumental measurement uncertainty is used in a **Type B evaluation of measurement uncertainty**.

Note 3 to entry: Information relevant to instrumental measurement uncertainty may be given in the instrument specifications.

4.25.
(5.19) accuracy class
class of **measuring instruments** or **measuring systems** that meet stated metrological requirements that are intended to keep **measurement**
errors or **instrumental measurement uncertainties** within specified limits under specified operating conditions

Note 1 to entry: An accuracy class is usually denoted by a number or symbol adopted by convention.

Note 2 to entry: Accuracy class applies to **material measures**.

4.26.
(5.21) maximum permissible measurement error
maximum permissible error limit of error

extreme value of **measurement error**, with respect to a known **reference quantity value**, permitted by specifications or regulations for a given **measurement**, **measuring instrument**, or **measuring system**

Note 1 to entry: Usually, the term “maximum permissible errors” or “limits of error” is used where there are two extreme values.

Note 2 to entry: The term “tolerance” should not be used to designate ‘maximum permissible error’.

4.27.
(5.22) datum measurement error
datum error

measurement error of a **measuring instrument** or **measuring system** at a specified **measured quantity value**

4.28.
(5.23) zero error
datum measurement error where the specified **measured quantity value** is zero

Note 1 to entry: Zero error should not be confused with absence of **measurement error**.

4.29.
null measurement uncertainty
measurement uncertainty where the specified **measured quantity value** is zero

Note 1 to entry: Null measurement uncertainty is associated with a null or near zero **indication** and covers an interval where one does not know whether the **measurand** is too small to be detected or the indication of the **measuring instrument** is due only to noise.

Note 2 to entry: The concept of ‘null measurement uncertainty’ also applies when a difference is obtained between **measurement** of a sample and a blank.

4.30.
calibration diagram
graphical expression of the relation between **indication** and corresponding **measurement result**

Note 1 to entry: A calibration diagram is the strip of the plane defined by the axis of the indication and the axis of measurement result, that represents the relation between an indication and a set of **measured quantity values** . A one-to-many relation is given, and the width of the strip for a given indication provides the **instrumental measurement uncertainty**.

Note 2 to entry: Alternative expressions of the relation include a **calibration curve** and associated **measurement uncertainty**, a calibration table, or a set of functions.

Note 3 to entry: This concept pertains to a **calibration** when the instrumental measurement uncertainty is large in comparison with the measurement uncertainties associated with the **quantity values** of **measurement standards** .

4.31.
calibration curve
expression of the relation between **indication** and corresponding **measured quantity value**

Note 1 to entry: A calibration curve expresses a one-to-one relation that does not supply a **measurement result** as it bears no information about the **measurement uncertainty**.

5. Measurement standards (Etalons)

For the purposes of this document, the following terms and definitions apply.

5.1.
(6.1) measurement standard
etalon

realization of the definition of a given **quantity**, with stated **quantity value** and associated **measurement uncertainty**, used as a reference
Frequently, this component is small compared with other components of the combined standard measurement uncertainty.

Note 1 to entry: A “realization of the definition of a given quantity” can be provided by a **measuring system**, a **material measure**, or a reference material.

Note 2 to entry: A measurement standard is frequently used as a reference in establishing **measured quantity values** and associated measurement uncertainties for other quantities of the same **kind**, thereby establishing **metrological traceability** through **calibration** of other measurement standards, **measuring instruments**, or measuring systems.

Note 3 to entry: The term “realization” is used here in the most general meaning. It denotes three procedures of “realization”. The first one consists in the physical realization of the **measurement unit** from its definition and is realization *sensu stricto*. The second, termed “reproduction”, consists not in realizing the measurement unit from its definition but in setting up a highly reproducible measurement standard based on a physical phenomenon, as it happens, e.g. in case of use of frequency-stabilized lasers to establish a measurement standard for the metre, of the Josephson effect for the volt or of the quantum Hall effect for the ohm. The third procedure consists in adopting a material measure as a measurement standard. It occurs in the case of the measurement standard of 1 kg.

Note 4 to entry: A standard measurement uncertainty associated with a measurement standard is always a component of the **combined standard measurement uncertainty** (see [Guide 1995], 2.3.4) in a **measurement result** obtained using the measurement standard.

Note 5 to entry: Quantity value and measurement uncertainty must be determined at the time when the measurement standard is used.

Note 6 to entry: Several quantities of the same kind or of different kinds may be realized in one device which is commonly also called a measurement standard.

Note 7 to entry: The word “embodiment” is sometimes used in the English language instead of “realization”.

Note 8 to entry: In science and technology, the English word “standard” is used with at least two different meanings: as a specification, technical recommendation, or similar normative document (in French “norme”) and as a measurement standard (in French “étalon”). This Vocabulary is concerned solely with the second meaning.

Note 9 to entry: The term “measurement standard” is sometimes used to denote other metrological tools, e.g. ‘software measurement standard’ (see [ISO 5436-2]).

EXAMPLE 1 1 kg mass measurement standard with an associated **standard measurement uncertainty** of 3 µg.

EXAMPLE 2 100 measurement standard resistor with an associated standard measurement uncertainty of 1µ.

EXAMPLE 3 Caesium frequency standard with a relative standard measurement uncertainty of $2 \cdot 10^{-15}$.

EXAMPLE 4 Standard buffer solution with a pH of 7.072 with an associated standard measurement uncertainty of 0.006.

EXAMPLE 5 Set of reference solutions of cortisol in human serum having a certified quantity value with measurement uncertainty for each solution.

EXAMPLE 6 **Reference material** providing quantity values with measurement uncertainties for the mass concentration of each of ten different proteins.

5.2.
(6.2) international measurement standard
measurement standard recognized by signatories to an international agreement and intended to serve worldwide

EXAMPLE 1 The international prototype of the kilogram.

EXAMPLE 2 Chorionic gonadotrophin, World Health Organization (WHO) 4th international standard 1999, 75/589, 650 International Units per ampoule.

EXAMPLE 3 VSMOW2 (Vienna Standard Mean Ocean Water) distributed by the International Atomic Energy Agency (IAEA) for differential stable isotope amount-of-substance ratio **measurements**.

5.3.
(6.3) national measurement standard
national standard

measurement standard recognized by national authority to serve in a state or economy as the basis for assigning **quantity values** to other measurement standards for the **kind of quantity** concerned

5.4.
(6.4) primary measurement standard
primary standard

measurement standard established using a **primary reference measurement procedure**, or created as an artifact, chosen by convention

EXAMPLE 1 Primary measurement standard of amount-of-substance concentration prepared by dissolving a known amount of substance of a chemical component to a known volume of solution.

EXAMPLE 2 Primary measurement standard for pressure based on separate **measurements** of force and area.

EXAMPLE 3 Primary measurement standard for isotope amount-of-substance ratio measurements, prepared by mixing known amount-of-substances of specified isotopes.

EXAMPLE 4 Triple-point-of-water cell as a primary measurement standard of thermodynamic temperature.

EXAMPLE 5 The international prototype of the kilogram as an artifact, chosen by convention.

5.5.
(6.5) secondary measurement standard
secondary standard

measurement standard established through **calibration** with respect to a **primary measurement standard** for a **quantity** of the same **kind**

Note 1 to entry: Calibration may be obtained directly between a primary measurement standard and a secondary measurement standard, or involve an intermediate **measuring system** calibrated by the primary measurement standard and assigning a **measurement result** to the secondary measurement standard.

Note 2 to entry: A measurement standard having its **quantity value** assigned by a ratio **primary reference measurement procedure** is a secondary measurement standard.

5.6.
(6.6) reference measurement standard
reference standard

measurement standard designated for the **calibration** of other measurement standards for **quantities** of a given **kind** in a given organization or at a given location

5.7.
(6.7) working measurement standard
working standard

measurement standard that is used routinely to calibrate or verify **measuring instruments** or **measuring systems**

Note 1 to entry: A working measurement standard is usually calibrated with respect to a **reference measurement standard**.

Note 2 to entry: In relation to **verification**, the terms “check standard” or “control standard” are also sometimes used.

5.8.
(6.9) travelling measurement standard
travelling standard

measurement standard, sometimes of special construction, intended for transport between different locations

EXAMPLE Portable battery-operated caesium-133 frequency measurement standard.

5.9.
(6.8) transfer measurement device
transfer device

device used as an intermediary to compare **measurement standards**

Note 1 to entry: Sometimes, measurement standards are used as transfer devices.

5.10.
intrinsic measurement standard
intrinsic standard

measurement standard based on an inherent and reproducible property of a phenomenon or substance

Note 1 to entry: A **quantity value** of an intrinsic measurement standard is assigned by consensus and does not need to be established by relating it to another measurement standard of the same type. Its **measurement uncertainty** is determined by considering two components: the first associated with its consensus quantity value and the second associated with its construction, implementation, and maintenance.

Note 2 to entry: An intrinsic measurement standard usually consists of a system produced according to the requirements of a consensus procedure and subject to periodic **verification**. The consensus procedure may contain provisions for the application of **corrections** necessitated by the implementation.

Note 3 to entry: Intrinsic measurement standards that are based on quantum phenomena usually have outstanding **stability**.

Note 4 to entry: The adjective “intrinsic” does not mean that such a measurement standard may be implemented and used without special care or that such a measurement standard is immune to internal and external influences.

EXAMPLE 1 Triple-point-of-water cell as an intrinsic measurement standard of thermodynamic temperature.

EXAMPLE 2 Intrinsic measurement standard of electric potential difference based on the Josephson effect.

EXAMPLE 3 Intrinsic measurement standard of electric resistance based on the quantum Hall effect.

EXAMPLE 4 Sample of copper as an intrinsic measurement standard of electric conductivity.

5.11.
(6.12) conservation of a measurement standard
maintenance of a measurement standard

set of operations necessary to preserve the metrological properties of a **measurement standard** within stated limits

Note 1 to entry: Conservation commonly includes periodic **verification** of predefined metrological properties or **calibration**, storage under suitable conditions, and specified care in use.

5.12.
calibrator
measurement standard used in **calibration**

Note 1 to entry: The term “calibrator” is only used in certain fields.

5.13.
(6.13) reference material
RM

material, sufficiently homogeneous and stable with reference to specified properties, which has been established to be fit for its intended use in **measurement** or in examination of **nominal properties**

Note 1 to entry: Examination of a nominal property provides a nominal property value and associated uncertainty. This uncertainty is not a **measurement uncertainty**.

Note 2 to entry: Reference materials with or without assigned **quantity values** can be used for **measurement precision** control whereas only reference materials with assigned quantity values can be used for **calibration** or **measurement trueness** control.

Note 3 to entry: ‘Reference material’ comprises materials embodying **quantities** as well as nominal properties.

Note 4 to entry: A reference material is sometimes incorporated into a specially fabricated device.

Note 5 to entry: Some reference materials have assigned quantity values that are metrologically traceable to a **measurement unit** outside a **system of units**. Such materials include vaccines to which International Units (IU) have been assigned by the World Health Organization.

Note 6 to entry: In a given **measurement**, a given reference material can only be used for either calibration or quality assurance.

Note 7 to entry: The specifications of a reference material should include its material traceability, indicating its origin and processing (Accred. Qual. Assur.:2006).

Note 8 to entry: ISO/REMCO has an analogous definition [EMONS 2006] but uses the term “measurement process” to mean ‘examination’ ([ISO 15189:2007], 3.4), which covers both measurement of a quantity and examination of a nominal property.

EXAMPLE 1 *Examples of reference materials embodying quantities*:

1. water of stated purity, the dynamic viscosity of which is used to calibrate viscometers;
2. human serum without an assigned quantity value for the amount-of-substance concentration of the inherent cholesterol, used only as a measurement precision control material;
3. fish tissue containing a stated mass fraction of a dioxin, used as a **calibrator**.

EXAMPLE 2 *Examples of reference materials embodying nominal properties*:

1. colour chart indicating one or more specified colours;
2. DNA compound containing a specified nucleotide sequence;
3. urine containing 19-androstenedione.

EXAMPLE 3 Substance of known triple-point in a triple-point cell.

EXAMPLE 4 Glass of known optical density in a transmission filter holder.

EXAMPLE 5 Spheres of uniform size mounted on a microscope slide.

5.14.
(6.14) certified reference material
CRM

reference material, accompanied by documentation issued by an authoritative body and providing one or more specified property values with associated uncertainties and traceabilities, using valid procedures

Note 1 to entry: ‘Documentation’ is given in the form of a ‘certificate’ (see [ISO Guide 31:2000]).

Note 2 to entry: Procedures for the production and certification of certified reference materials are given, e.g. in [ISO Guide 34:2000] and [ISO Guide 35:2006].

Note 3 to entry: In this definition, “uncertainty” covers both ‘measurement uncertainty’ and ‘uncertainty associated with the value of a **nominal property**‘, such as for identity and sequence. “Traceability” covers both ‘ **metrological traceability** of a quantity value’ and ‘traceability of a nominal property value’.

Note 4 to entry: Specified quantity values of certified reference materials require metrological traceability with associated measurement uncertainty (Accred. Qual. Assur.:2006).

Note 5 to entry: ISO/REMCO has an analogous definition (Accred. Qual. Assur.:2006) but uses the modifiers “metrological” and “metrologically” to refer to both quantity and nominal property.

EXAMPLE Human serum with assigned **quantity value** for the concentration of cholesterol and associated **measurement uncertainty** stated in an accompanying certificate, used as a **calibrator** or **measurement trueness** control material.

5.15.
commutability of a reference material
property of a **reference material**, demonstrated by the closeness of agreement between the relation among the **measurement results** for a stated **quantity** in this material, obtained according to two given **measurement procedures**, and the relation obtained among the measurement results for other specified materials

Note 1 to entry: The reference material in question is usually a **calibrator** and the other specified materials are usually routine samples.

Note 2 to entry: The measurement procedures referred to in the definition are the one preceding and the one following the reference material (calibrator) in question in a **calibration hierarchy** (see [ISO 17511]).

Note 3 to entry: The stability of commutable reference materials should be monitored regularly.

5.16.
reference data
data related to a property of a phenomenon, body, or substance, or to a system of components of known composition or structure, obtained from an identified source, critically evaluated, and verified for accuracy

Note 1 to entry: In this definition, accuracy covers, for example, **measurement accuracy** and ‘accuracy of a nominal property value’.

Note 2 to entry: “Data” is a plural form, “datum” is the singular. “Data” is commonly used in the singular sense, instead of “datum”.

EXAMPLE Reference data for solubility of chemical compounds as published by the IUPAC.

5.17.
standard reference data
reference data issued by a recognized authority

EXAMPLE 1 Values of the fundamental physical constants, as regularly evaluated and recommended by CODATA of ICSU.

EXAMPLE 2 Relative atomic mass values, also called atomic weight values, of the elements, as evaluated every two years by IUPAC-CIAAW, approved by the IUPAC General Assembly, and published in *Pure Appl. Chem*.

5.18.
reference quantity value
reference value

quantity value used as a basis for comparison with values of **quantities** of the same **kind**

Note 1 to entry: A reference quantity value can be a **true quantity value** of a **measurand**, in which case it is unknown, or a **conventional quantity value**, in which case it is known.

Note 2 to entry: A reference quantity value with associated **measurement uncertainty** is usually provided with reference to

1. a material, e.g. a **certified reference material**,
2. a device, e.g. a stabilized laser,
3. a **reference measurement procedure**,
4. a comparison of **measurement standards**.

6. List of acronyms

BIPM	International Bureau of Weights and Measures
CCQM	Consultative Committee for Amount of Substance — Metrology in Chemistry

CGPM	General Conference on Weights and Measures
CODATA	Committee on Data for Science and Technology
GUM	Guide to the Expression of Uncertainty in Measurement
IAEA	International Atomic Energy Agency
ICSU	International Council for Science
IEC	International Electrotechnical Commission
IFCC	International Federation of Clinical Chemistry and Laboratory Medicine
ILAC	International Laboratory Accreditation Cooperation
ISO	International Organization for Standardization
ISO/REMCO	International Organization for Standardization, Reference Materials Committee
IUPAC	International Union of Pure and Applied Chemistry
IUPAC/CIAAW	International Union of Pure and Applied Chemistry — Commission on Isotopic Abundances and Atomic Weights
IUPAP	International Union of Pure andApplied Physics
JCGM	Joint Committee for Guides in Metrology
JCGM/WG 1	Joint Committee for Guides in Metrology, Working Group 1 on the GUM
JCGM/WG 2	Joint Committee for Guides in Metrology, Working Group 2 on the VIM
OIML	International Organization of Legal Metrology
VIM, 2nd edition	International Vocabulary of Basic and General Terms in Metrology (1993)
VIM, 3rd edition	International Vocabulary of Metrology — Basic and General Concepts and Associated Terms (this publication) VIM
VIML	International Vocabulary of Terms in Legal Metrology
WHO	World Health Organization

Annex A
Concept diagrams

The 12 concept diagrams in this informative Annex are intended to provide:

— a visual presentation of the relations between the concepts defined and termed in the preceding clauses;

— a possibility for checking whether the definitions offer adequate relations;

— a background for identifying further needed concepts; and

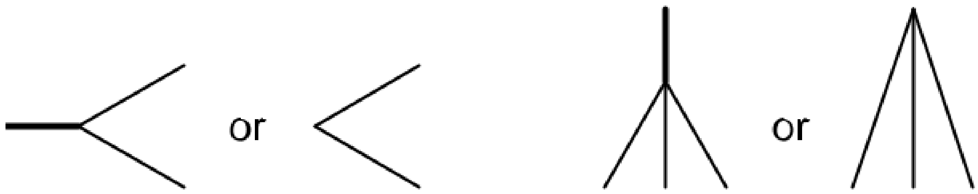
— a check that terms are sufficiently systematic.

It should be recalled, however, that a given concept may be describable by many characteristics and only essential delimiting characteristics are included in the definition.

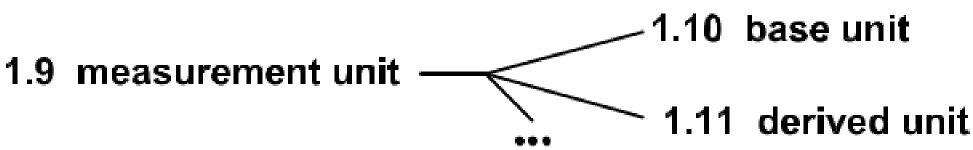
The area available on a page limits the number of concepts that can be presented legibly, but all diagrams are in principle interrelated as indicated in each diagram by parenthetic references to other diagrams.

The relations used are of three types as defined by [ISO 704:2000] and [ISO 1087-1:2000]. Two are hierarchical, i.e. having superordinate and subordinate concepts, the third is non-hierarchical.

The hierarchical *generic relation* (or genus-species relation) connects a generic concept and a specific concept; the latter inherits all characteristics of the former. The diagrams show such relations as a tree,

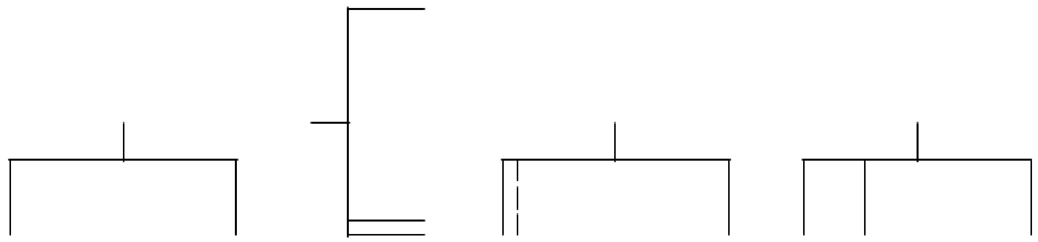


where a short branch with three dots indicates that one or more other specific concepts exist, but are not included for presentation and a heavy starting line of a tree shows a separate terminological dimension. For example,

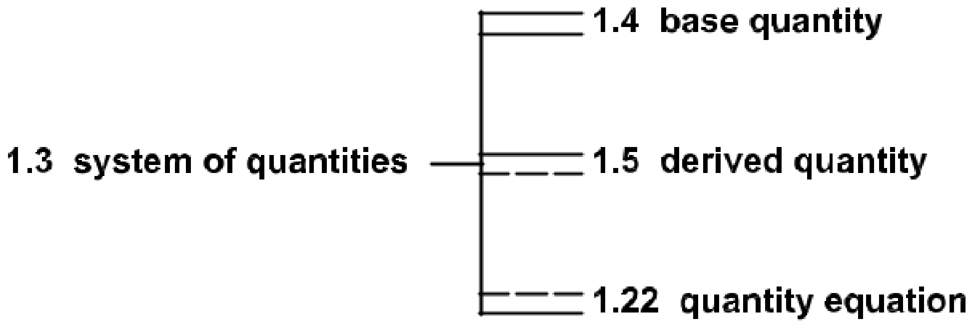


where a third concept might be ‘off-system measurement unit’.

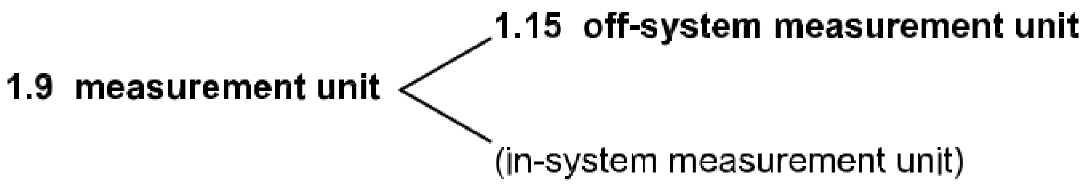
The *partitive relation* (or part-whole relation) is also hierarchical and connects a comprehensive concept to two or more partitive concepts which fitted together constitute the comprehensive concept. The diagrams show such relations as a rake or bracket, and a continued backline without a tooth means one or more further partitive concepts that are not discussed.



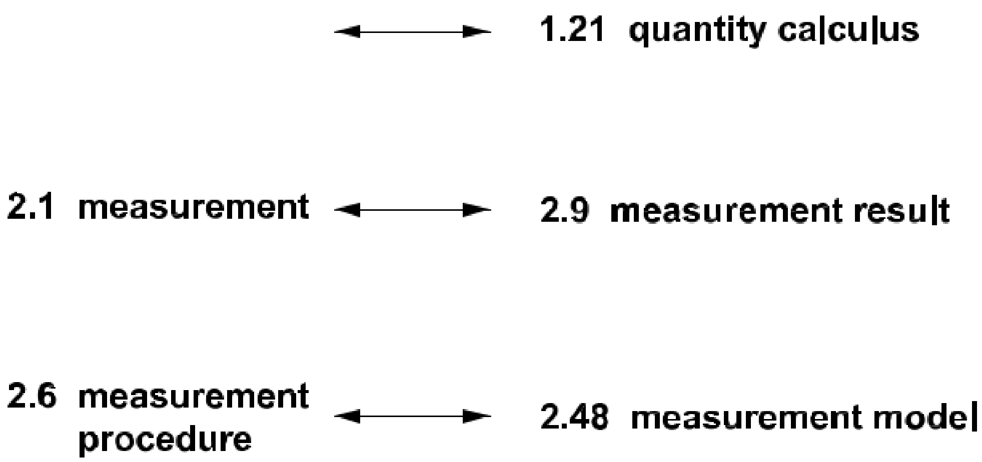
A close-set double line indicates that several partitive concepts of a given type are involved and a broken line shows that such plurality is uncertain. For example



A parenthetic term indicates a concept that is not defined in the Vocabulary, but is taken as a primitive which is assumed to be generally understood.



The *associative relation* (or pragmatic relation) is non-hierarchical and connects two concepts which are in some sort of thematic association. There are many subtypes of associative relation, but all are indicated by a double-headed arrow. For example,



To avoid too complicated diagrams, they do not show all the possible associative relations. The diagrams will demonstrate that fully systematic derived terms have not been created, often because metrology is an old discipline with a vocabulary evolved by accretion rather than as a comprehensive de novo structure.

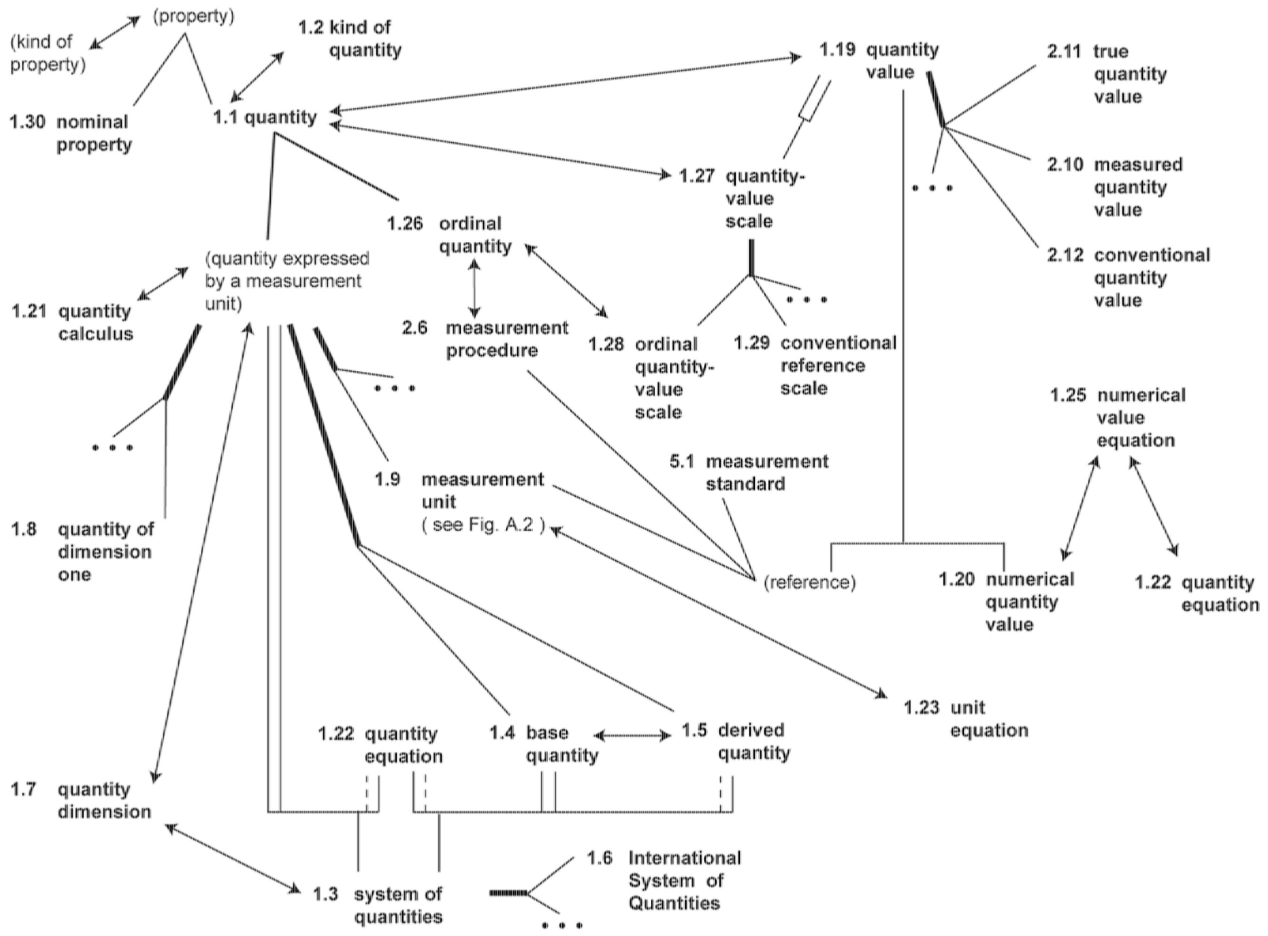


Figure A.1 — Concept diagram for part of Clause 1 around 'quantity'

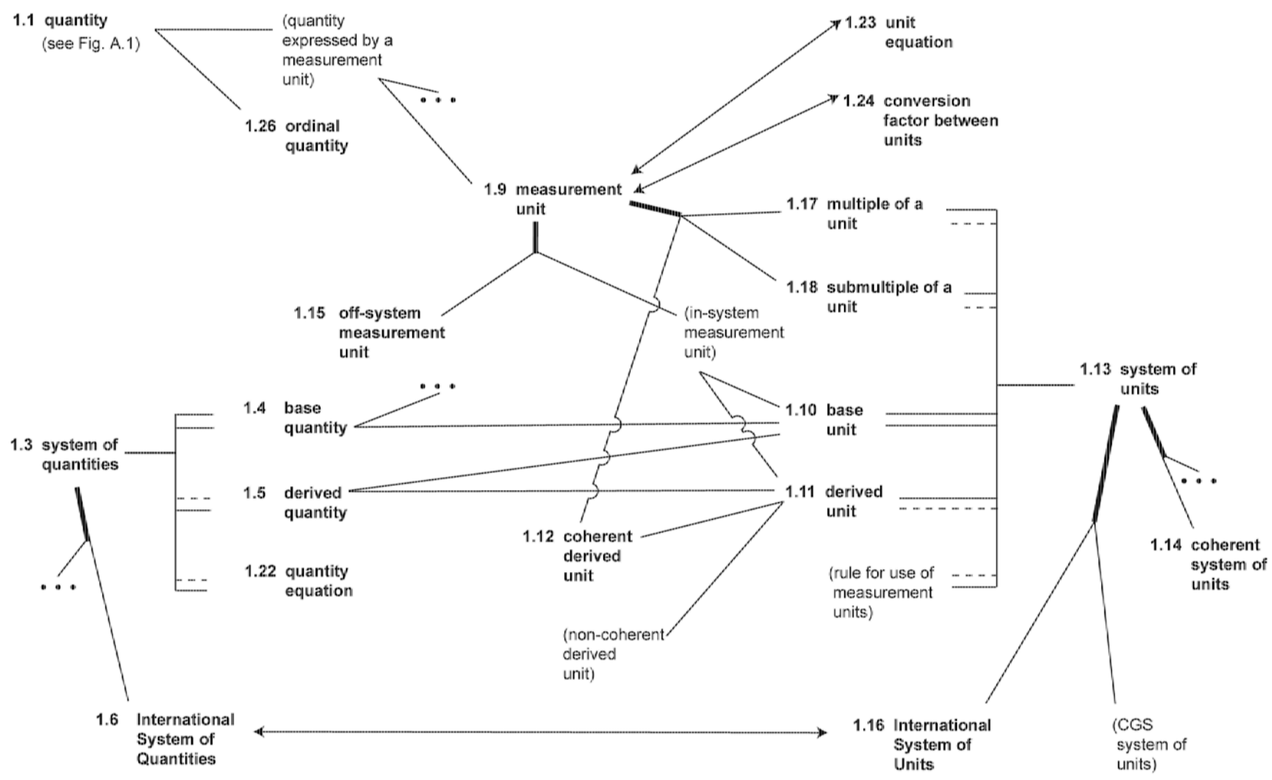


Figure A.2 — Concept diagram for part of Clause 1 around 'measurement unit'

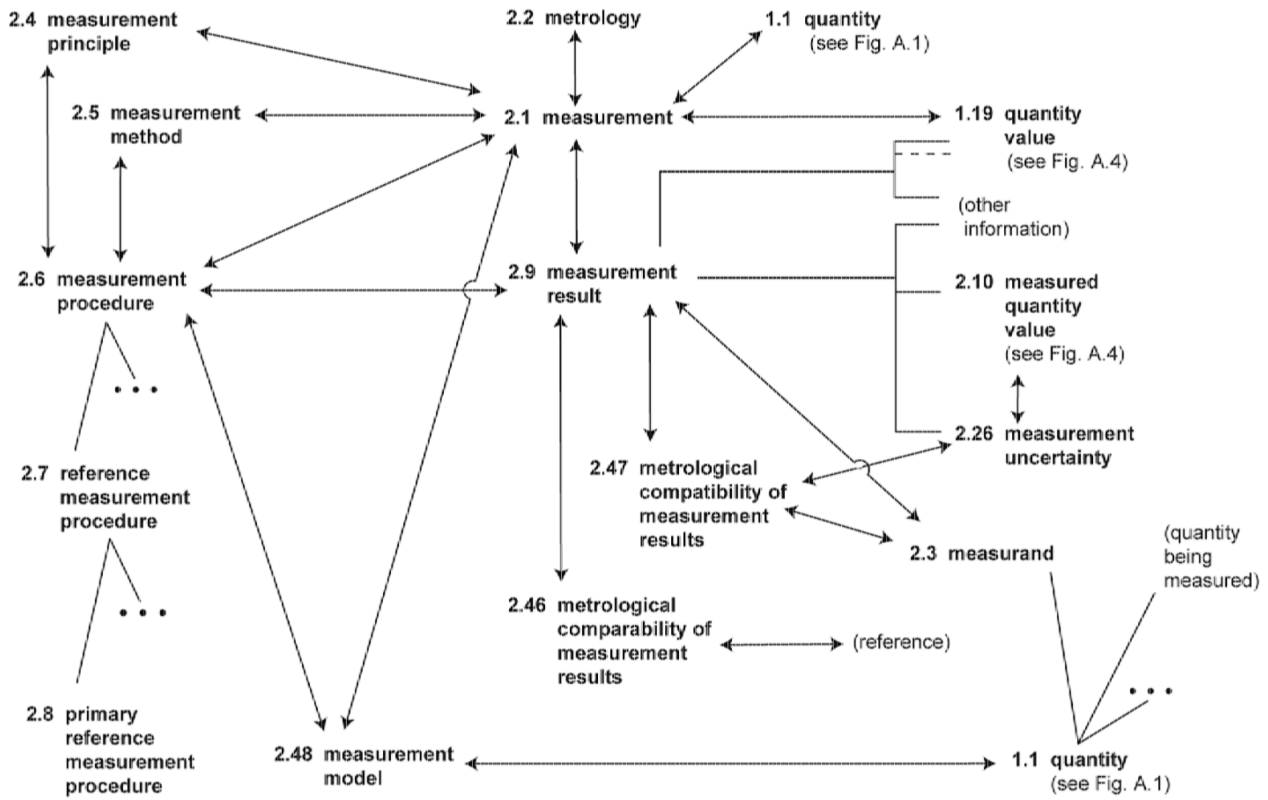


Figure A.3 — Concept diagram for part of Clause 2 around 'measurement'

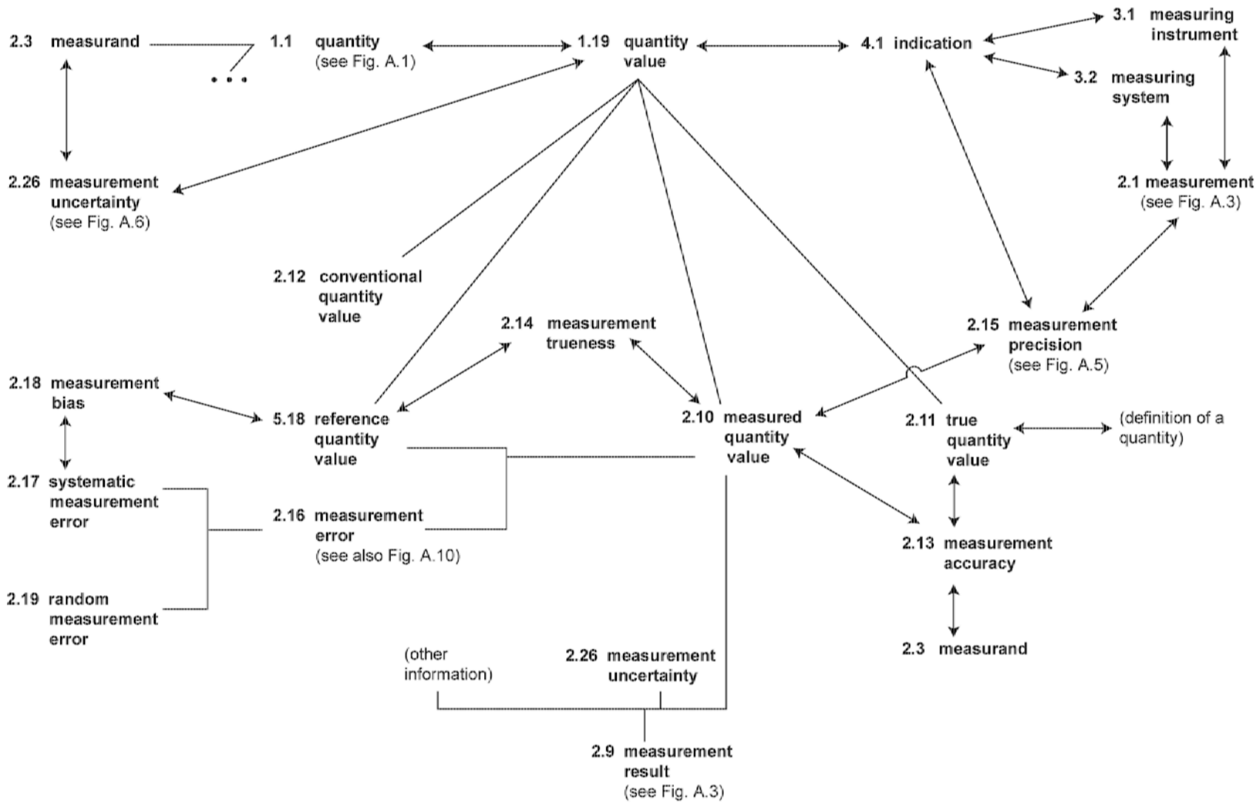


Figure A.4 — Concept diagram for part of Clause 2 around 'quantity value'

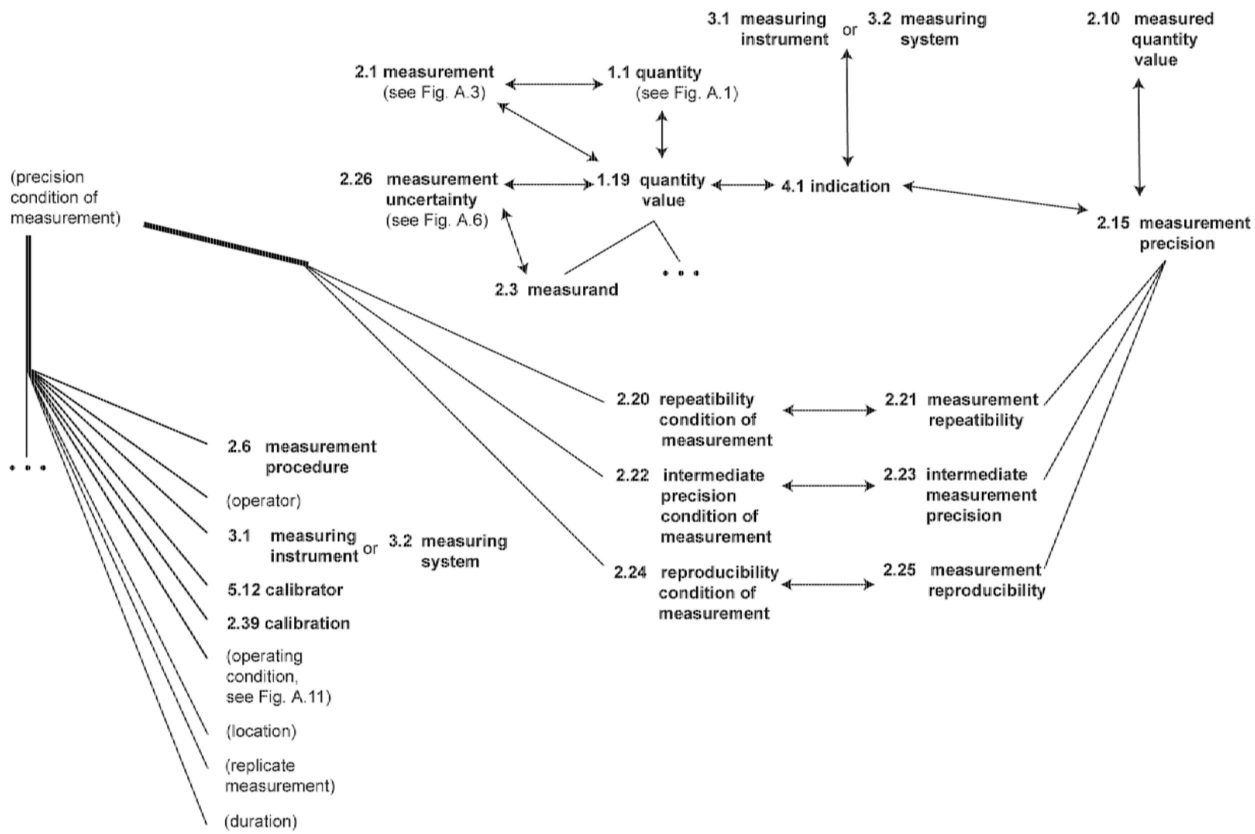


Figure A.5 — Concept diagram for part of Clause 2 around 'measurement precision'

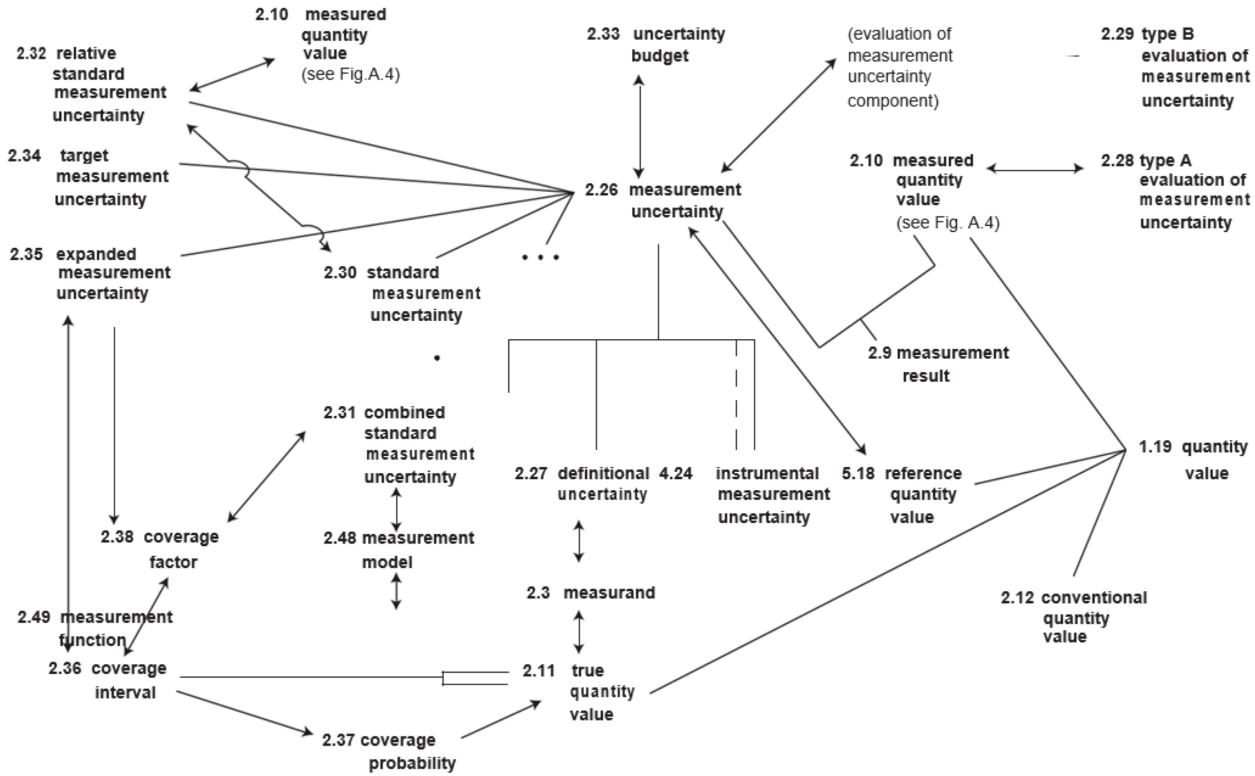


Figure A.6 — Concept diagram for part of Clause 2 around 'measurement uncertainty'

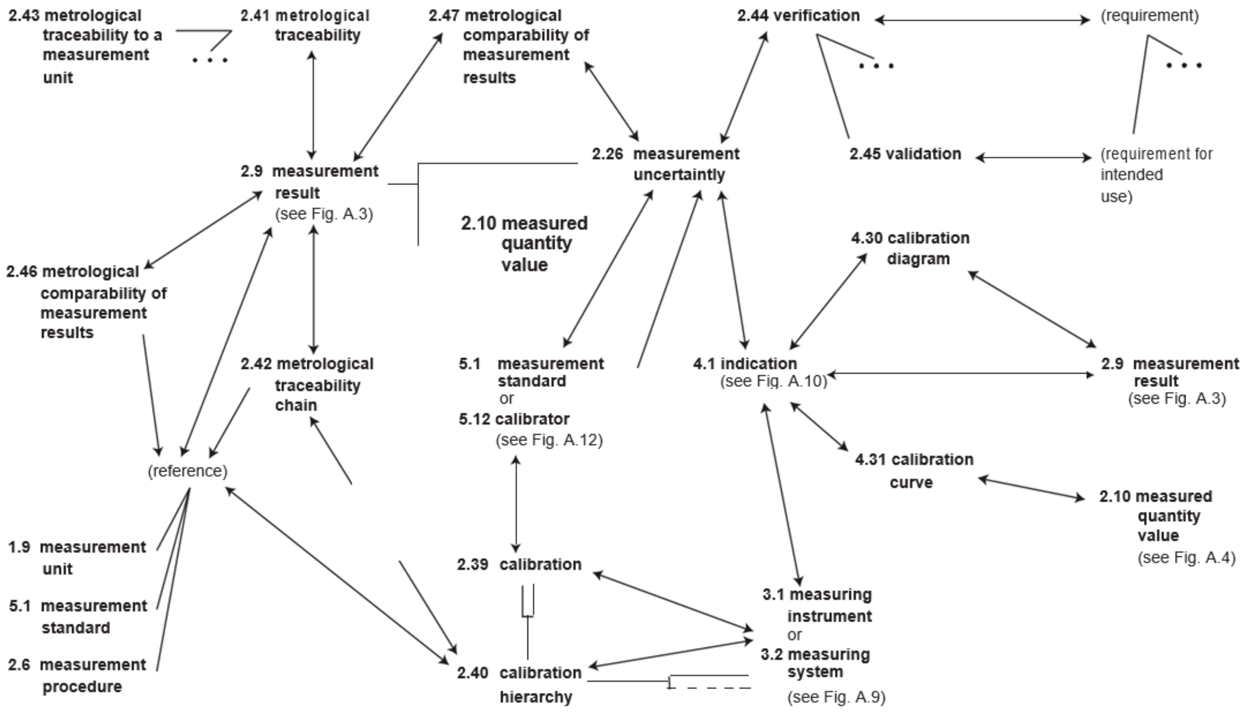


Figure A.7 — Concept diagram for part of Clause 2 around 'calibration'

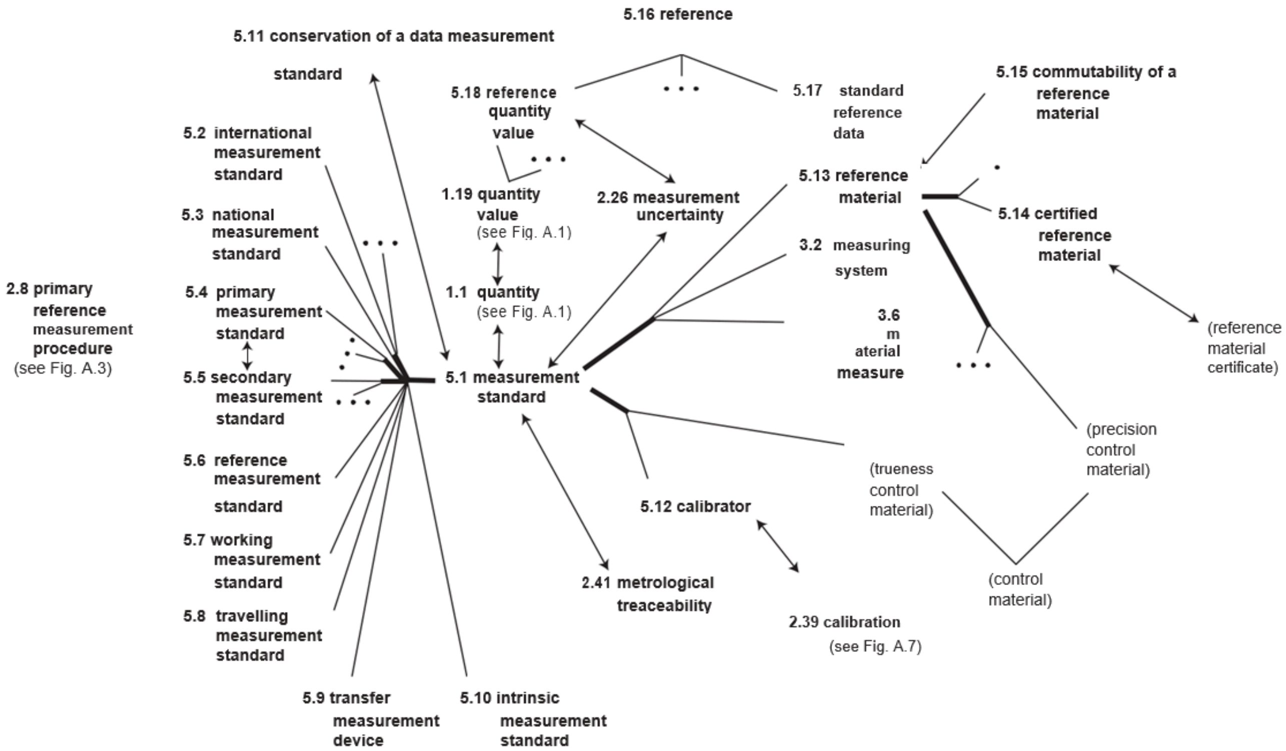


Figure A.8 — Concept diagram for part of Clause 5 around 'measurement

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